

Modelling Financial Markets

Chapter 11: Continuous-Time Models



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How do we describe prices in continuous time?

What the models explain

- Prices are recorded daily or every few milliseconds: can one model describe them at **every** time scale at once?
- Part I: Brownian motion, Itô calculus, simulation, geometric Brownian motion (GBM), mean reversion
- Part II: estimating diffusions, change of measure, jumps, stochastic volatility, models against data, a case study, AI-assisted discovery

What you should be able to do

- apply Itô's lemma and simulate a stochastic differential equation with a known error order
- estimate GBM, Ornstein–Uhlenbeck, Merton and Heston on data, with standard errors that respect what the sample identifies
- say which feature of the data each model reproduces, and which feature none of them does
- explain why option pricing (Chapter 12) and interest-rate models use this language

Indices and Data Sources

- **VIX**: Cboe Volatility Index.
- **BET**: Bucharest Exchange Trading (index).
- **BVB**: Bursa de Valori București — the Bucharest Stock Exchange.
- **CBOE**: Chicago Board Options Exchange.
- **FRED**: Federal Reserve Economic Data.
- **DTB3**: 3-Month Treasury Bill: secondary market rate (FRED series code).

Data Used in This Chapter

Series	Type	Period	Observations / frequency
S&P 500	US equity index	1990–2026	9 245
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
VIX	CBOE implied volatility index of the S&P 500	1990–2026	daily
DTB3	3-month US Treasury bill rate (FRED)	1954–2026	daily

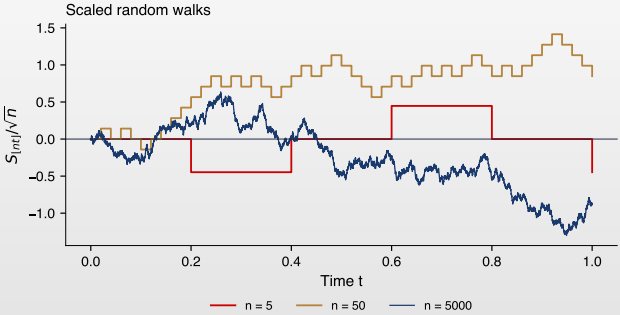
- Daily closing values up to 18 September 2026; daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar (cleaning rules: Appendix)
- Time in years, daily step $\Delta t = 1/252$ (Bitcoin: $1/365$); drifts and volatilities annualised
- Intraday: SPY (exchange-traded fund, ETF, tracking the S&P 500) 5-minute bars, October 2020 – September 2026, 1 468 full trading days (case study; Chapter 9)
- Notation
  - W_t is standard Brownian motion and S_t (also P_t) the price
  - v_t is the instantaneous variance and $\mathcal{F}_t$ the information (filtration) up to t
  - $\mathrm{d}S/S$ is the instantaneous simple return and $\mathrm{d}\ln S$ the log change; Itô's lemma explains their different drifts

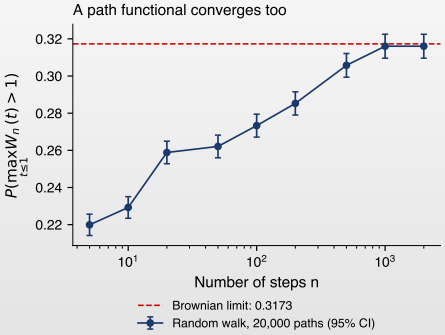
From Random Walk to Brownian Motion: Donsker's Theorem (1/3)

- **Question**: prices move in discrete steps; which process do we get as the steps become smaller and more frequent?
- **Random walk**: $S_0 = 0$, $S_k = S_{k-1} + \xi_k = \xi_1 + \cdots + \xi_k$, with steps ξ_i i.i.d. (independent and identically distributed), mean 0 and variance 1
  - It is the log-price model of Chapter 2, with ξ_k the standardised return of day k
- **Scaled random walk**: $W_n(t) = S_{\lfloor nt \rfloor}/\sqrt{n}$, with n steps per unit of time ($\lfloor nt \rfloor$: integer part of nt)
  - Why $\sqrt{n}$: $\mathrm{Var}(W_n(t)) = \lfloor nt \rfloor / n \to t$; divided by n the limit would be 0, undivided the variance would explode
- Central limit theorem (CLT): $W_n(t) \to N(0, t)$ for each t separately
- Donsker's theorem: $W_n \Rightarrow W$ ($\Rightarrow$: convergence in distribution) for the **whole path**, whatever the law of the steps Billingsley (1999)

From Random Walk to Brownian Motion: Donsker's Theorem (2/3)

CI: Confidence Interval.





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From Random Walk to Brownian Motion: Donsker's Theorem (3/3)

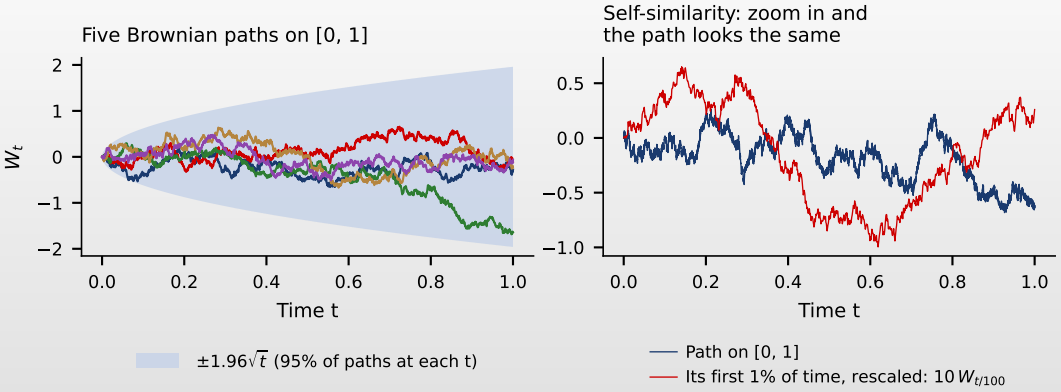
- Left: three scaled random walks on $[0,1]$ with $n = 5$, 50 and 5 000 steps; the vertical axis is $S_{\lfloor nt \rfloor} / \sqrt{n}$
  - For large n the step path becomes continuous and irregular, like a Brownian path
- Right: $P(\max_{t \le 1} W_n(t) > 1)$, estimated from 20 000 paths (95% CI), against n on a log axis
  - The probability rises from 0.220 ($n = 5$) to 0.316 ($n = 2000$), towards the reflection-principle value $2(1 - \Phi(1)) = 0.3173$ (Φ: standard Normal distribution function)
- A coarse grid misses the maximum reached between observations
- Convergence is slow, of order $1/\sqrt{n}$: daily closes understate the intraday range

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Brownian Motion: Definition and Properties

- **Standard Brownian motion**: $W_0 = 0$; independent increments; $W_t - W_s \sim N(0, t-s)$ for $s < t$; continuous paths
  - Existence: Wiener (1923); simulation on a grid: $W_{t+\Delta t} = W_t + \sqrt{\Delta t}\, Z$, $Z \sim N(0,1)$
- A Gaussian process with $\mathrm{Cov}(W_s, W_t) = \min(s, t)$
- A **martingale**: $\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s$, with $\mathcal{F}_s$ the information up to s (the efficient-market random walk of Chapter 2)
- **Self-similarity**: $\{c^{-1/2} W_{ct}\}$ is again a standard Brownian motion for any $c > 0$
- Paths are continuous but **nowhere differentiable**
  - $(W_{t+h} - W_t)/h$ has standard deviation $1/\sqrt{h} \to \infty$: ordinary calculus fails

Brownian Paths (1/2)



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Brownian Paths (2/2)

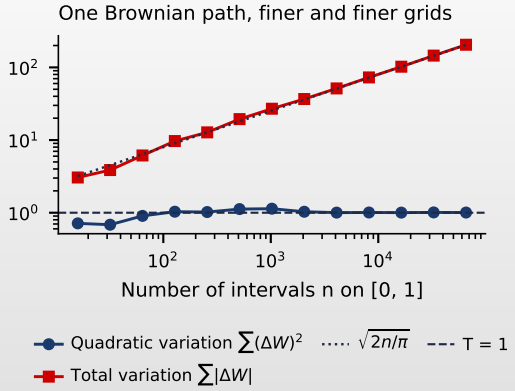
- Left: five paths with the pointwise band $\pm 1.96\sqrt{t}$, which covers 95% for each t separately, not for the whole path
- Right: a path and its first 1% of time, stretched 100 times in time and 10 times in space
- The band widens like $\sqrt{t}$ (the square-root-of-time rule of Chapter 7)
- A zoomed piece looks like the whole path
  - Real markets break self-similarity: fat tails and clustering depend on the horizon (Chapter 1)

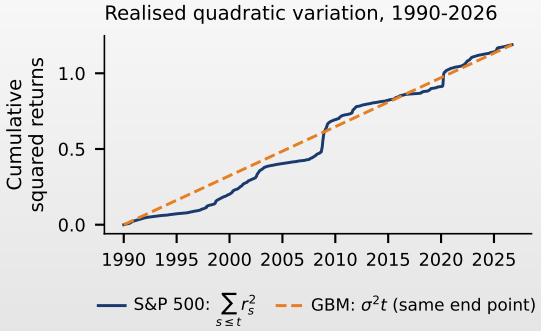
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Total and Quadratic Variation

- **Question**: why does ordinary calculus fail on a Brownian path, and what replaces it?
- **Total variation** $\sum_k |\Delta W_k|$, on the grid $0 = t_0 < \cdots < t_n = T$ with step $\Delta t = T/n$ and $\Delta W_k = W_{t_k} - W_{t_{k-1}}$
  - Its mean is $\sqrt{2nT/\pi} \to \infty$: a Brownian path has infinite length
- **Quadratic variation** $[W]_T = \lim \sum_k (\Delta W_k)^2 = T$: a random sum with a **deterministic** limit
  - $Q_n = \sum_{k=1}^{n} (\Delta W_k)^2$, $\mathrm{Var}(Q_n) = 2T^2/n$: one year of daily data gives about 9% relative error ($\sqrt{2/252}$), the idea behind realised variance (Chapter 9)
- Shorthand of Itô calculus: $(\mathrm{d}W_t)^2 = \mathrm{d}t$

Quadratic Variation: Simulation and Data (1/2)





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Quadratic Variation: Simulation and Data (2/2)

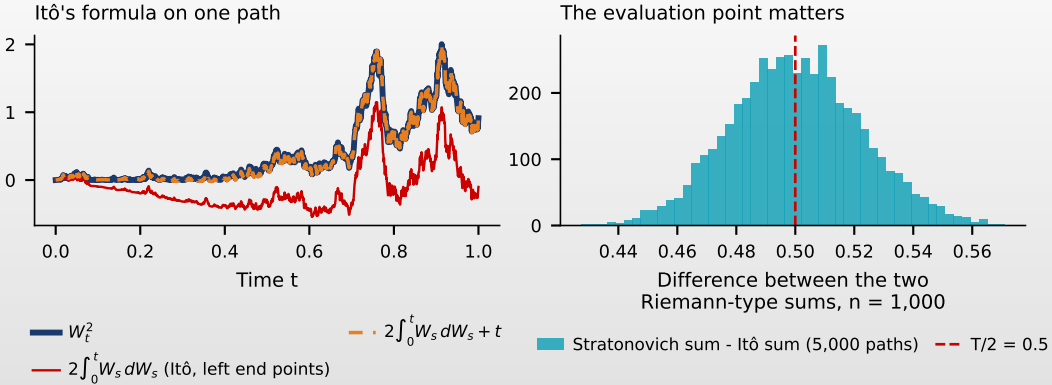
- Left: one Brownian path on grids with 16 to 65,536 intervals: the quadratic variation settles at 1.006 ($T = 1$), while the total variation keeps growing, 205 against $\sqrt{2n/\pi} = 204$
- Right: cumulative $\sum r_t^2$ of the S&P 500, 1.188 over 36.7 years (average volatility 18.0%); under geometric Brownian motion it would follow the straight line $\sigma^2 t$
- The sum rises in steps
  - September 2008 – March 2009: 1.6% of the days, but 15.2% of the quadratic variation
  - February – April 2020: 0.5% of the days, but 7.8% of the quadratic variation
  - volatility is itself random (Heston, Section 10)

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The Itô Integral

- Goal: a meaning for $\int_0^T H_t \, \mathrm{d}W_t$, the gain of holding H_t units of an asset priced W; Riemann–Stieltjes integration fails because the total variation is infinite
- **Itô integral**: the limit in mean square of sums with the integrand at the **left** end point
  - $\int_0^T H_t \, \mathrm{d}W_t = \lim_{n\to\infty} \sum_{k=1}^{n} H_{t_{k-1}} \left(W_{t_k} - W_{t_{k-1}}\right)$, for H **adapted** with $\mathbb{E}\left[\int_0^T H_t^2 \, \mathrm{d}t\right] < \infty$
  - H_t uses only information up to t: the position is chosen before the next price move
- Zero mean, $\mathbb{E}\left[\int_0^T H_t \, \mathrm{d}W_t\right] = 0$, and the **Itô isometry**, $\mathbb{E}\left[\left(\int_0^T H_t \, \mathrm{d}W_t\right)^2\right] = \mathbb{E}\left[\int_0^T H_t^2 \, \mathrm{d}t\right]$
  - $t \mapsto \int_0^t H_s \, \mathrm{d}W_s$ is a martingale: no strategy earns money on average (weaker conditions: Appendix)
- The integrand averaged over the end points, $\frac{1}{2}(H_{t_{k-1}} + H_{t_k})$, gives the **Stratonovich** integral, a different number; finance uses Itô, because that position would use the price at t_k

Worked Example: $\int_0^T W_t \, \mathrm{d}W_t$ on Simulated Paths (1/2)



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Worked Example: $\int_0^T W_t \, \mathrm{d}W_t$ on Simulated Paths (2/2)

- Each term: $W_{t_{k-1}} \Delta W_k = \frac{1}{2}(W_{t_k}^2 - W_{t_{k-1}}^2) - \frac{1}{2}(\Delta W_k)^2$; summed over k, the first parts give $\frac{1}{2}(W_T^2 - W_0^2) = \frac{1}{2} W_T^2$, because the intermediate terms cancel, and the second parts tend to $\frac{1}{2} T$: $\int_0^T W_t \, \mathrm{d}W_t = \frac{1}{2} W_T^2 - \frac{1}{2} T$
  - Ordinary calculus would give $\frac{1}{2} W_T^2$: the extra $-\frac{1}{2} T$ is the quadratic variation, and the mean is $\frac{1}{2}(T - T) = 0$
- Left: on one path of 1 000 steps, $2 \int_0^t W \, \mathrm{d}W + t$ lies exactly on W_t^2
- Right: Stratonovich minus Itô sums over 5,000 paths have mean 0.500 and standard deviation 0.022; theory gives $T/2 = 0.5$
- Itô sums have mean 0.007 (theory: 0) and variance 0.520 (theory, by the isometry: $\int_0^1 t \, \mathrm{d}t = 1/2$)

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Itô Processes and Itô's Lemma

- **Itô process**: $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$ (drift a_t, diffusion coefficient b_t, both adapted)
  - Shorthand for $X_t = X_0 + \int_0^t a_s\,\mathrm{d}s + \int_0^t b_s\,\mathrm{d}W_s$
- **Itô's lemma**: for $f(t,x)$ continuously differentiable once in t and twice in x Itô (1944)
  - $\mathrm{d}f(t,X_t) = \left(f_t + a_t f_x + \frac{1}{2}b_t^2 f_{xx}\right)\mathrm{d}t + b_t f_x\,\mathrm{d}W_t$
  - Partial derivatives evaluated at (t,X_t): $f_t = \partial f/\partial t$, $f_x = \partial f/\partial x$, $f_{xx} = \partial^2 f/\partial x^2$
- Rule of thumb: Taylor expansion to second order with $\mathrm{d}t\cdot\mathrm{d}t = 0$, $\mathrm{d}t\cdot\mathrm{d}W = 0$, $\mathrm{d}W\cdot\mathrm{d}W = \mathrm{d}t$
  - The second-order term $\frac{1}{2}b_t^2 f_{xx}\,\mathrm{d}t$ does not vanish: this is the whole novelty

Quadratic Covariation and the Multivariate Itô Formula

- **Quadratic covariation** $[X, Y]_t = \lim \sum_k \Delta X_k \, \Delta Y_k$; if $\mathrm{Corr}(\mathrm{d}W^{(1)}, \mathrm{d}W^{(2)}) = \rho$, then $\mathrm{d}[W^{(1)}, W^{(2)}]_t = \rho \, \mathrm{d}t$
  - Product rule: $\mathrm{d}(X_t Y_t) = X_t \, \mathrm{d}Y_t + Y_t \, \mathrm{d}X_t + \mathrm{d}[X, Y]_t$
- **Multivariate Itô formula** for $X_t \in \mathbb{R}^d$ and $f(t, x)$ continuously differentiable once in t and twice in x
  - $\mathrm{d}f(t, X_t) = f_t \, \mathrm{d}t + \sum_i f_{x_i} \, \mathrm{d}X_t^i + \frac{1}{2} \sum_{i,j} f_{x_i x_j} \, \mathrm{d}[X^i, X^j]_t$, with X_t^i the i-th component
- Example (Heston, Section 10): $\mathrm{d}S = \mu S \, \mathrm{d}t + \sqrt{v} \, S \, \mathrm{d}W^{(1)}$, $\mathrm{d}v = \kappa(\theta - v) \, \mathrm{d}t + \xi \sqrt{v} \, \mathrm{d}W^{(2)}$
  - v is the variance of the price S, κ the speed of mean reversion, θ the long-run variance and ξ the volatility of the variance
  - $\mathrm{d}(S_t v_t) = S_t \, \mathrm{d}v_t + v_t \, \mathrm{d}S_t + \rho \, \xi \, v_t S_t \, \mathrm{d}t$: ρ enters the drift of the product

Stochastic Differential Equations and the Euler–Maruyama Scheme

- **Question**: how do we simulate a model with no exact solution (Heston), and how large is the discretisation error?
- A **stochastic differential equation** (SDE): $\mathrm{d}X_t = a(t, X_t)\,\mathrm{d}t + b(t, X_t)\,\mathrm{d}W_t$, X_0 given: drift and diffusion depend on the current state
  - GBM, Ornstein–Uhlenbeck and Merton have exact transitions: we simulate them with no discretisation error
  - Heston has only a closed-form characteristic function
  - Existence conditions: ▸ Appendix
- **Euler–Maruyama**: freeze a and b over each step Higham (2001)
  - $X_{k+1} = X_k + a(X_k)\,\Delta t + b(X_k)\,\Delta W_k$, $\Delta W_k = \sqrt{\Delta t}\,Z_k$, with X_k the approximation at $t_k = k\Delta t$ and Z_k i.i.d. $N(0,1)$

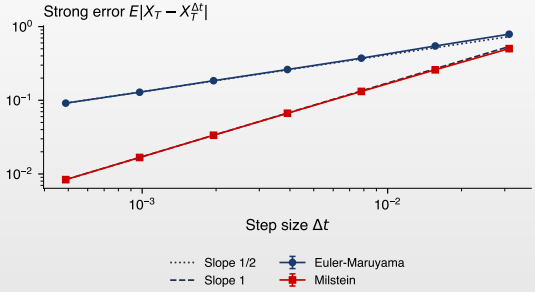
The Milstein Scheme and Strong Convergence

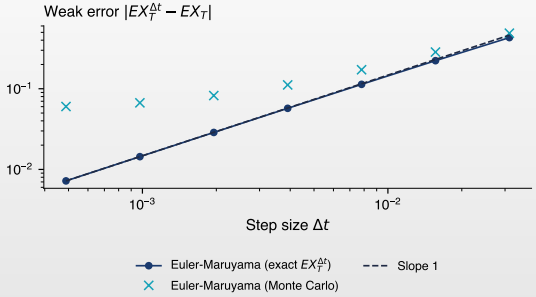
- **Milstein**: add the next Itô–Taylor term Milstein (1975)
  - $X_{k+1} = X_k + a(X_k)\,\Delta t + b(X_k)\,\Delta W_k + \frac{1}{2} b(X_k)\,b'(X_k)\left[(\Delta W_k)^2 - \Delta t\right]$, with $b' = \mathrm{d}b/\mathrm{d}x$
  - The correction has conditional mean zero: it raises the strong order (path accuracy), not the weak order
- **Strong error**: $\mathbb{E}|X_T - X_T^{\Delta t}| \leq C\,\Delta t^{\gamma}$, how close each simulated path is to the true one
  - X_T is the exact solution, $X_T^{\Delta t}$ the scheme with step Δt, C a constant and γ the strong order
  - $\gamma = 1/2$ for Euler–Maruyama and $\gamma = 1$ for Milstein; it matters for path-dependent quantities (hedging errors, barrier crossings)

Weak Convergence

- **Weak error**: $|\mathbb{E}[g(X_T^{\Delta t})] - \mathbb{E}[g(X_T)]| \leq C\,\Delta t^{\beta}$, how close the simulated distribution is to the true one
  - g is a function of the final value (e.g. an option pay-off) and β the weak order
  - Both schemes have $\beta = 1$, enough for European option prices and other expected values
  - Conditions: smooth coefficients with bounded derivatives and a smooth g Kloeden & Platen (1992)
  - Square-root coefficients (CIR (Cox–Ingersoll–Ross), Heston), quantiles such as Value-at-Risk (VaR) and barrier crossings can converge more slowly
- Test equation Higham (2001): $\mathrm{d}X = \lambda X\,\mathrm{d}t + \mu X\,\mathrm{d}W$, $\lambda = 2$, $\mu = 1$, $X_0 = 1$, $T = 1$, with the exact solution as benchmark
  - On a log-log plot, the order is the slope of the error against the step (next slide)

Convergence of the Schemes (1/2)





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Convergence of the Schemes (2/2)

- 20,000 Brownian paths on a fine grid of 2^{11} steps, coarsened to steps from 2^{-11} to 2^{-5}
  - Strong orders (slope of the log-log fit, 95% CI by resampling whole paths): Euler–Maruyama 0.518 $[0.512, 0.525]$, Milstein 0.986 $[0.984, 0.987]$
  - The narrow CIs measure Monte Carlo noise only; the small gap to 1/2 and 1 is finite-step curvature, not a different order
  - Halving Δt halves the Milstein error, but cuts the Euler error only by 29%
- Weak order of Euler–Maruyama from the exact mean $(1 + \lambda \Delta t)^{T/\Delta t}$: 0.985, a deterministic slope
  - The Monte Carlo estimates flatten at small steps: their standard error, about 0.068, exceeds the bias
  - The sample size, not the step, is often the binding constraint Glasserman (2003)

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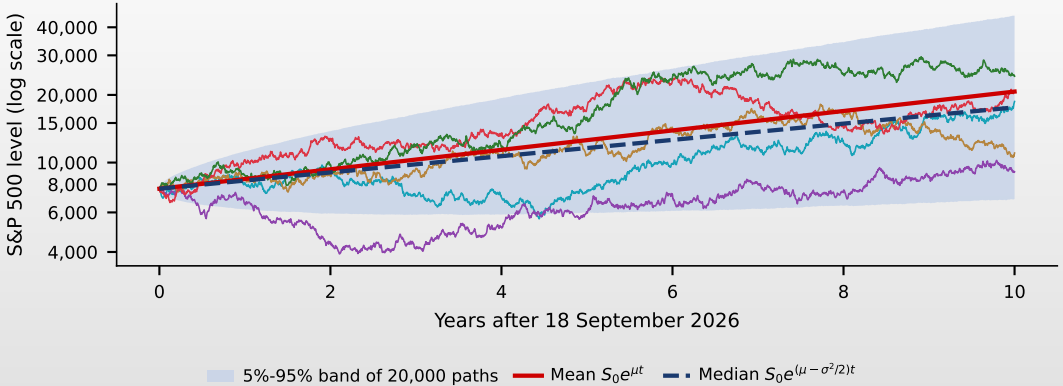
Geometric Brownian Motion (GBM) from Itô's Lemma (1/2)

- **Question**: which price model stays positive and gives percentage returns with the same distribution at every price level?
- SDE: $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$: the **percentage** change has constant drift μ and volatility σ per year
  - Proposed by Osborne (1959) and Samuelson (1965); the price model behind Black & Scholes (1973)
- Itô's lemma for $f(x) = \ln x$, with $f_x = 1/x$, $f_{xx} = -1/x^2$, $a_t = \mu S_t$, $b_t = \sigma S_t$
  - $\mathrm{d}\ln S_t = \left(\mu - \frac{1}{2}\sigma^2\right)\mathrm{d}t + \sigma\,\mathrm{d}W_t$, so $S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right)$
  - Prices stay positive; exact simulation needs only Normal draws

Geometric Brownian Motion (GBM) from Itô's Lemma (2/2)

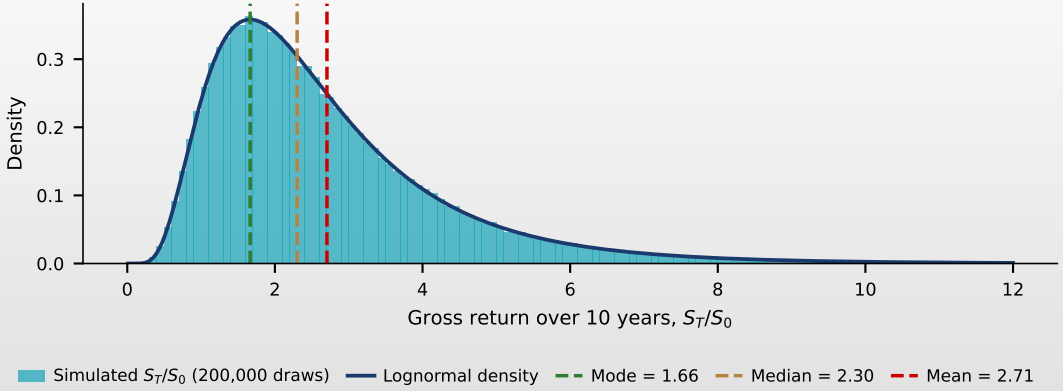
- The log price grows at the log drift $m = \mu - \sigma^2/2$, not at μ: volatility lowers the growth of the median investor's wealth
  - With $\sigma = 18\%$, the cost is $\sigma^2/2 = 1.6$ percentage points a year
  - With $\sigma = 67\%$ (Bitcoin), the cost reaches 22 percentage points a year
- Testable implications: $r_t = \ln(S_t/S_{t-\Delta t})$ i.i.d. $N(m\Delta t, \sigma^2\Delta t)$ at every frequency
  - Skewness and excess kurtosis are zero
  - There is no autocorrelation in r_t, $|r_t|$ or r_t^2
  - The variance is proportional to the horizon

Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths (1/3)



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Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths (2/3)



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Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths (3/3)

- GBM with $\hat{\mu} = 10.0\%$ and $\hat{\sigma} = 18.0\%$ (1990–2026), over $T = 10$ years: mean $\mathbb{E}[S_T] = S_0 e^{\mu T}$, median $S_0 e^{mT}$, mode $S_0 e^{(m-\sigma^2)T}$
  - Density (200 000 draws): the gross return S_T/S_0 is 2.71 at the mean, 2.30 at the median and 1.66 at the mode
- Fan chart: 20,000 exact paths starting from $S_0 = 7\,650$ (18 September 2026)
  - 90% of the paths end between 6 878 (5% quantile) and 45 017 (95% quantile)
  - Probability of a 10-year loss: $P(S_T < S_0) = \Phi(-m\sqrt{T}/\sigma) = 7.1\%$
- $P(S_T < \mathbb{E}[S_T]) = \Phi(\sigma\sqrt{T}/2) = 61.2\%$: most investors end below the average, so report the median and quantiles
- The fan is too narrow in crises: GBM has no fat tails and no volatility clustering (tested below)

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Estimating GBM by Maximum Likelihood

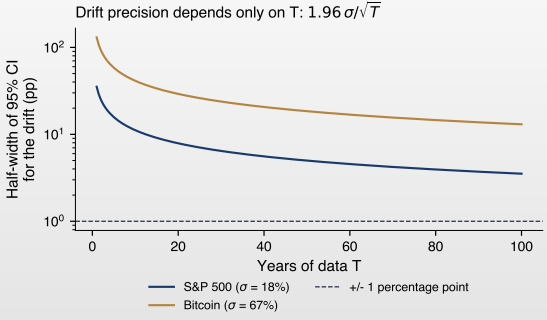
- Maximum likelihood estimation (MLE) from n log returns at step Δt, over $T = n\Delta t$ years ($\bar{r}$: mean log return): $\hat{\sigma}^2 = \frac{1}{n\Delta t}\sum_t (r_t - \bar{r})^2$, $\hat{m} = \bar{r}/\Delta t$, $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2$
  - Under GBM, the standard error (SE) $\mathrm{SE}(\hat{\sigma}) = \sigma/\sqrt{2n}$ shrinks with the number of observations
  - $\mathrm{SE}(\hat{m}) = \sigma/\sqrt{T}$ depends only on the **calendar span**, since $\sum_t r_t = \ln(S_T/S_0)$ uses only the first and the last price

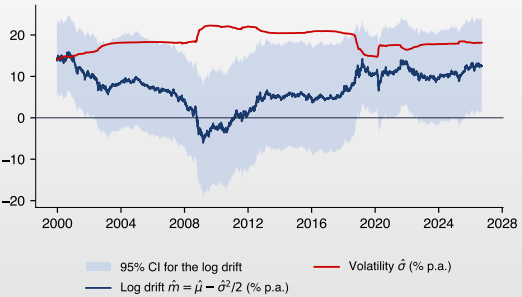
	Years	$\hat{\sigma}$ (%)	$\mathrm{SE}(\hat{\sigma})$	$\hat{m}$ (%)	$\mathrm{SE}(\hat{m})$
S&P 500	36.7	18.0	0.13	8.3	3.0
BET	26.5	22.2	0.19	16.1	4.3
Bitcoin	12.0	66.6	0.71	43.1	19.2

- The SE is 0.1–0.7 percentage points for volatility, but 3–19 points for the log drift
- S&P 500: $\hat{\mu} = 8.3\% + 1.62\% = 10.0\%$ per year
- These SE are conditional on GBM: with excess kurtosis k, even independent returns inflate $\mathrm{SE}(\hat{\sigma})$ by $\sqrt{(k+2)/2} = 2.5$ (S&P 500); block-bootstrap SE: Seminar 11, B1

How Precisely Can We Estimate the Drift? (1/2)

p.p. = percentage points.





MFM_ch11_drift_precision

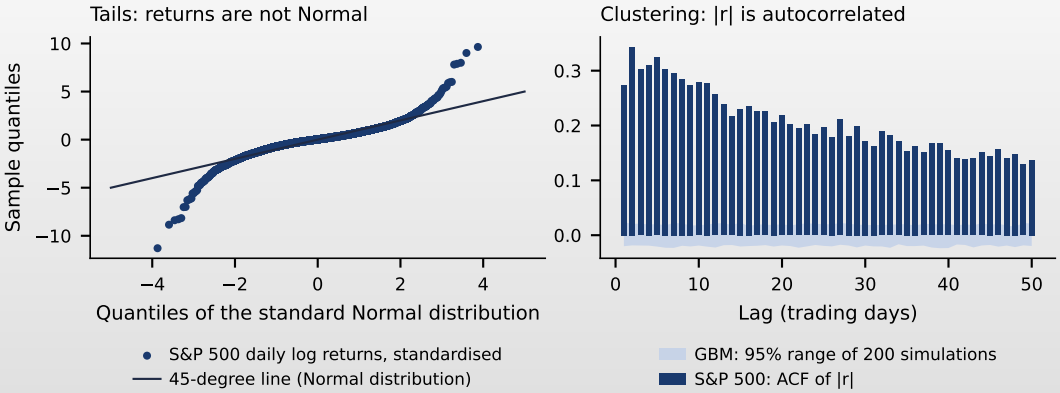
How Precisely Can We Estimate the Drift? (2/2)

- Left: half-width of the 95% CI for the drift against years of data
- A precision of ± 1 point needs $(1.96\,\hat{\sigma}/0.01)^2 = 1\,243$ years for the S&P 500 and 17 028 years for Bitcoin
- Right: S&P 500 on rolling 10-year windows
  - The log drift ranges from -6.0% (window ending 2009-03) to 15.8% (2000-09), each with a CI of about ± 11 points, so the variation is mostly noise
  - Volatility stays between 14.1% and 22.3%: precise, but not constant
- Option pricing in Chapter 12 avoids μ: under the risk-neutral measure only σ enters (Girsanov, Section 8)

MFM_ch11_drift_precision

Does GBM Fit the S&P 500? (1/2)

ACF: AutoCorrelation Function.



MFM_ch11_gbm_vs_data

Does GBM Fit the S&P 500? (2/2)

- Left: quantile–quantile (QQ) plot of standardised daily log returns against the Normal distribution
  - Excess kurtosis 10.9 instead of 0; worst day at -11.3 standard deviations
  - 62 days beyond ± 4 standard deviations, against 0.59 expected under the Normal distribution
  - The Jarque–Bera (JB) statistic, which tests zero skewness and zero excess kurtosis jointly, is 45 805 against the 5% critical value 5.99 (χ^2_2)
- Right: autocorrelation function (ACF) of $|r_t|$: 0.27 at lag 1 and still 0.14 at lag 50
  - The 95% band from 200 GBM samples of the same length stays below 0.024 in absolute value
- GBM fails on both stylised facts of Chapter 1 Cont (2001); the rest of the chapter adds mean reversion, jumps and random volatility

MFM_ch11_gbm_vs_data

Recap: Growth, Lognormality and Drift Precision

- Formulas to keep
  - $\ln(S_T/S_0) \sim N(mT, \sigma^2 T)$ with $m = \mu - \frac{1}{2}\sigma^2$: the median grows at m, the mean $\mathbb{E}[S_T]$ at μ
  - $\mathrm{SE}(\hat{\sigma}) = \sigma/\sqrt{2n}$ shrinks with the number of returns; $\mathrm{SE}(\hat{m}) = \sigma/\sqrt{T}$ only with the years covered
- Check: with $\sigma = 20\%$, what is the SE of the log drift after 10 years of data, and after 40 years?
  - Answer: $0.20/\sqrt{10} = 6.3$ and $0.20/\sqrt{40} = 3.2$ percentage points; four times the span only halves the error, whatever the sampling frequency

The Ornstein–Uhlenbeck (OU) Process

- **Question**: how do we model a variable that returns to a normal level, such as an interest rate or the VIX, instead of drifting away like a stock price?
- SDE: $\mathrm{d}x_t = \kappa(\theta - x_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, $\kappa > 0$ Uhlenbeck & Ornstein (1930): long-run mean θ, speed of mean reversion κ, volatility σ
  - Above θ the drift is negative, below θ positive: a force pulling x back to θ
  - Used for interest rates Vasicek (1977), volatility, spreads and commodity prices
- **Half-life** $\ln 2/\kappa$: the time for the expected deviation from θ to halve
  - With $\kappa = 0.5$ per year, the half-life is 1.39 years

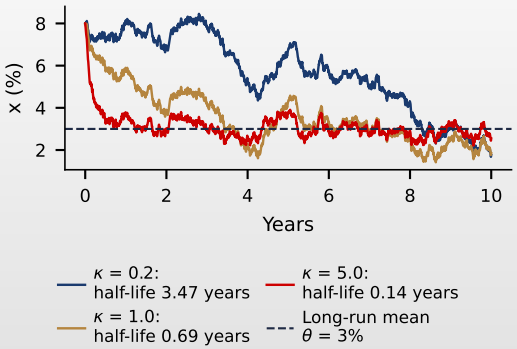
Solving the OU Equation in Three Steps

- Step 1: Itô's lemma on $f(t,x) = e^{\kappa t}x$ ($f_{xx} = 0$) gives $\mathrm{d}(e^{\kappa t}x_t) = \kappa\theta e^{\kappa t}\,\mathrm{d}t + \sigma e^{\kappa t}\,\mathrm{d}W_t$: the terms in x_t cancel
- Step 2: integrate from 0 to t and multiply by $e^{-\kappa t}$: $x_t = \theta + (x_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)}\,\mathrm{d}W_s$
- Step 3: the Itô integral has mean 0 and, by the isometry, variance $\sigma^2 \int_0^t e^{-2\kappa(t-s)}\,\mathrm{d}s$, so x_t is Normal with
  - $\mathbb{E}[x_t \mid x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$ and $\mathrm{Var}(x_t \mid x_0) = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right)$
  - Stationary distribution $N\left(\theta, \sigma^2/(2\kappa)\right)$: unlike Brownian motion, the variance stops growing with the horizon
- Check: $x_0 = 8\%$, $\theta = 3\%$, $\kappa = 0.5$, $\sigma = 1\%$: what are the mean and the standard deviation of x_1?
  - Answer: $3\% + 5\% \cdot e^{-0.5} = 6.03\%$ and $1\% \cdot \sqrt{1 - e^{-1}} = 0.80\%$: after one year 61% of the initial gap remains

Exact Discretisation: an AR(1)

- Sampled at step Δt, the OU process is exactly an autoregressive model of order 1 (AR(1))
  - $x_{t+\Delta t} = a + b\,x_t + \varepsilon_t$, $b = e^{-\kappa \Delta t}$, $a = \theta(1-b)$, $\mathrm{Var}(\varepsilon_t) = \frac{\sigma^2}{2\kappa}(1-b^2)$
- Maximum likelihood = ordinary least squares (OLS) of $x_{t+\Delta t}$ on x_t (intercept $\hat{a}$, slope $\hat{b}$, residual standard deviation $\hat{s}_\varepsilon$), then invert the mapping
  - $\hat{\kappa} = -\ln \hat{b}/\Delta t$, $\hat{\theta} = \hat{a}/(1-\hat{b})$, $\hat{\sigma} = \hat{s}_\varepsilon \sqrt{2\hat{\kappa}/(1-\hat{b}^2)}$
- Standard errors of $\hat{\kappa}$ and $\hat{\theta}$: the delta method, from the OLS covariance of $(\hat{a}, \hat{b})$
  - $\hat{\kappa}$ depends on $\hat{b}$ only: $\partial\kappa/\partial b = -1/(b\Delta t)$, so $\mathrm{Var}(\hat{b})$ suffices (Seminar 11, A1)
- With daily data b is very close to 1: κ is estimated imprecisely, as in unit-root tests

Three Speeds of Mean Reversion



- OU paths driven by the same shocks, starting at 8%
  - $\theta = 3\%$, $\sigma = 1\%$, and speed $\kappa = 0.2$ (slow), 1 or 5 (fast); exact simulation, daily steps
- Half-lives: 3.47 years for $\kappa = 0.2$, 0.69 for $\kappa = 1$ and 0.14 for $\kappa = 5$
  - With $\kappa = 5$ the gap from 8% to 3% closes within a few months
- Stationary standard deviations $\sigma/\sqrt{2\kappa}$: 1.58% for $\kappa = 0.2$, 0.71% for $\kappa = 1$ and 0.32% for $\kappa = 5$
  - Slow reversion looks like a random walk for years
- Long-horizon forecasts converge to θ

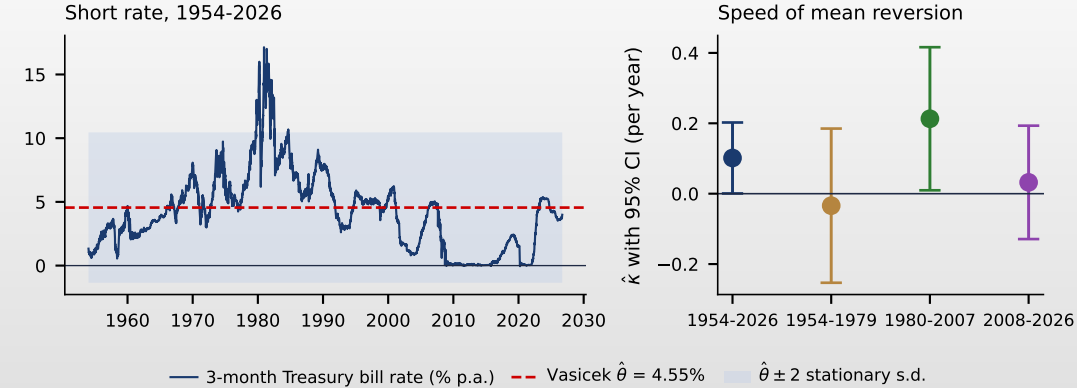
MFM_ch11_ou_paths

Short-Rate Models: Vasicek and CIR

- **Vasicek**: the short rate r_t is an OU process Vasicek (1977)
  - Normal distribution: negative rates have positive probability $\Phi(-\theta\sqrt{2\kappa}/\sigma)$ in the long run
  - Bond prices and yields in closed form (Feynman–Kac, Section 8)
- **Cox–Ingersoll–Ross** (CIR): $\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t$ Cox, Ingersoll & Ross (1985)
  - Volatility shrinks near zero
  - The rate stays positive if $2\kappa\theta \geq \sigma^2$ (Feller condition) Feller (1951)
- Which is closer to the data is an empirical question, tested in Section 7 Chan, Karolyi, Longstaff & Sanders (1992)

Vasicek on the 3-Month Treasury Bill Rate (1/2)

s.d. = standard deviation.



MFM_ch11_vasicek

Vasicek on the 3-Month Treasury Bill Rate (2/2)

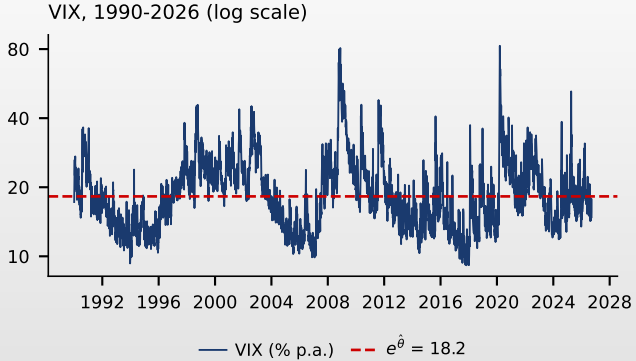
- Left: daily 3-month US Treasury bill rate (DTB3), 1954–2026, with the Vasicek long-run mean $\hat{\theta} = 4.55\%$ and the band $\hat{\theta}$ plus/minus two stationary standard deviations (s.d.)
  - $\hat{\kappa} = 0.101$ (SE 0.051), a half-life of 6.8 years; volatility $\hat{\sigma} = 1.33\%$
  - Daily AR(1) coefficient $\hat{b} = 0.99960$: almost a unit root
- Right: $\hat{\kappa}$ with 95% CI over three periods: -0.034 in 1954–1979 (explosive), 0.213 in 1980–2007 and 0.032 in 2008–2026
  - Two of the three CIs include zero
- Test of the random-walk null ($\kappa = 0$): parametric bootstrap with 500 random walks of the same length, because under $\kappa = 0$ the distribution of $\hat{\kappa}$ is not Normal and is shifted upwards
  - p-value $= 0.287$: the sample does not establish mean reversion (Seminar 11, B2)

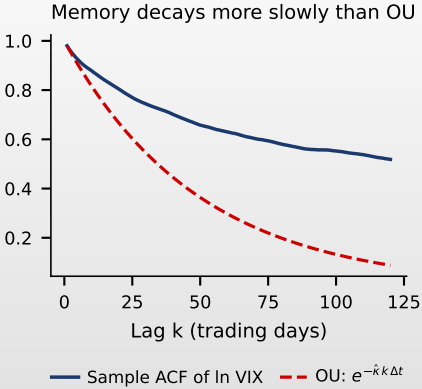
MFM_ch11_vasicek

Why Estimated Mean Reversion Looks Too Fast

- In finite samples $\hat{\kappa}$ overstates κ by roughly $4/T$ (T in years) Tang & Chen (2009)
  - Simulation: the full-sample estimates as the truth, 500 samples of 72 years
  - The mean $\hat{\kappa}$ is 0.164, a bias of 0.063, close to the approximation $4/T = 0.055$
  - 78% of the estimates exceed the truth (Seminar 11, B2)
- Where the bias comes from: $\hat{b}$, the OLS slope of $x_{t+\Delta t}$ on x_t, is biased towards zero in a short persistent sample, and $\hat{\kappa} = -\ln \hat{b}/\Delta t$ turns a too-small $\hat{b}$ into a too-large $\hat{\kappa}$
  - The bias depends on the span T, not on the number of observations: sampling daily instead of monthly does not remove it (Section 7)

Is Volatility Mean-Reverting? The VIX (1/2)





MFM_ch11_vix_ou

Is Volatility Mean-Reverting? The VIX (2/2)

- Left: VIX 1990–2026 on a log scale with the OU long-run level $e^{\hat{\theta}} = 18.2$
  - OU fit to $\ln \mathrm{VIX}$: $\hat{\kappa} = 5.10$ (SE 0.53), a half-life of 34 trading days
  - Spikes such as 82.7 (2020-03-16) fade within weeks
- Right: sample ACF of $\ln \mathrm{VIX}$ against the OU-implied $e^{-\hat{\kappa} k \Delta t}$
  - At lag 60, the autocorrelation is 0.63 in the data but 0.30 under OU: after the fast decay a slow component remains
- One OU factor is too simple: candidates are two-factor or long-memory models, while rough volatility (Section 12) is a distinct alternative
- Mean-reverting volatility is the building block of Heston

MFM_ch11_vix_ou

Recap: Mean Reversion in Three Formulas

- Formulas to keep
  - Expected path: $\mathbb{E}[x_t \mid x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$; half-life $\ln 2/\kappa$
  - Long-run spread: stationary standard deviation $\sigma/\sqrt{2\kappa}$
  - Estimation: AR(1) slope $b = e^{-\kappa \Delta t}$, hence $\hat{\kappa} = -\ln \hat{b}/\Delta t$, biased upwards in short samples
- Check: monthly data give $\hat{b} = 0.95$. What are $\hat{\kappa}$ and the half-life?
  - Answer: $\hat{\kappa} = -12 \ln 0.95 = 0.616$ per year and $\ln 2/0.616 = 1.13$ years, about 13.5 months

Infill Asymptotics: What Sampling More Often Buys

Question: which parameters do we learn from sampling more often, and which only from more years?

Infill: $\Delta \to 0$ on a fixed $[0, T]$

- $\sum_k (\Delta X_k)^2 \to \int_0^T b_t^2 \, \mathrm{d}t$
- Diffusion coefficient: identified without error in the limit
  - Quadratic variation is observable (continuous Itô process, no measurement noise)
- Sampling more often helps σ (Section 5) and $b(x)$ (nonparametric, below)

Long-Span Asymptotics: What Only More Years Buy

Long span: $T \to \infty$

- The drift is not consistently estimable on a fixed span
  - Fisher information $I(m) = -\mathbb{E}[\partial^2 \ell / \partial m^2]$ (ℓ: log-likelihood) bounds $\mathrm{Var}(\hat{m}) \geq 1/I(m)$; for n GBM returns $I(m) = n\Delta t/\sigma^2 = T/\sigma^2$, so $\mathrm{SE}(\hat{m}) = \sigma/\sqrt{T}$
- OU: $\mathrm{Var}(\hat{\kappa}) \approx (e^{2\kappa\Delta} - 1)/(T\Delta) \to 2\kappa/T$ as $\Delta \to 0$ (Seminar 11, A1)
- The bias of $\hat{\kappa}$ does not shrink with finer sampling Yu (2012)
  - $b = e^{-\kappa\Delta} \to 1$: $\hat{\kappa}$ is far from Normal when κT is small

- The imprecise GBM drift and the biased $\hat{\kappa}$ have the same cause: information grows with T, not with n

Likelihood for Nonlinear Diffusions (deeper)

- **Euler pseudo-likelihood**: the Gaussian transition $r_{t+\Delta} \mid r_t \sim N\left(r_t + a(r_t)\Delta,\ b^2(r_t)\Delta\right)$, with $a(r)$, $b(r)$ the drift and diffusion of the rate r; its bias, of order Δ, is small with daily data
- **Exact CIR**: $2c\, r_{t+\Delta} \mid r_t$ is noncentral χ^2 with $4\kappa\theta/\sigma^2$ degrees of freedom, $c = 2\kappa/\left(\sigma^2(1 - e^{-\kappa\Delta})\right)$ Cox, Ingersoll & Ross (1985)
  - The noncentrality parameter is $2c\, r_t e^{-\kappa\Delta}$
  - The Feller condition is equivalent to at least 2 degrees of freedom (Seminar 11, A7)
- General diffusions: closed-form Hermite expansions of the transition density Aït-Sahalia (2002)
- CKLS (Chan–Karolyi–Longstaff–Sanders): $\mathrm{d}r = \kappa(\theta - r)\,\mathrm{d}t + \sigma r^{\gamma}\,\mathrm{d}W$ nests Vasicek ($\gamma = 0$) and CIR ($\gamma = 1/2$) and is estimated by Gaussian (Euler) quasi-likelihood Chan, Karolyi, Longstaff & Sanders (1992)
  - Tests on γ use sandwich standard errors with a heteroskedasticity- and autocorrelation-consistent (HAC) score covariance Newey & West (1987)

Vasicek, CIR and CKLS on the T-Bill Rate, 1954–2007 (deeper)

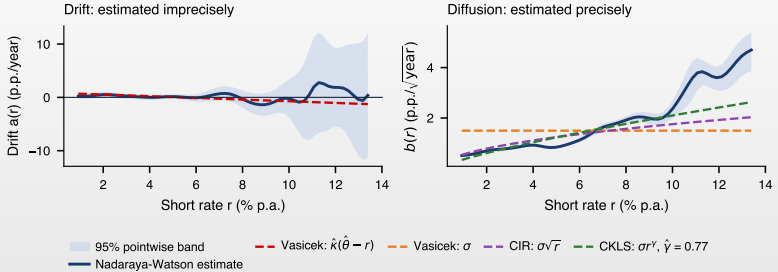
Model	$\hat{\kappa}$ daily (SE)	$\hat{\kappa}$ monthly (SE)	γ
Vasicek, exact	0.158 (0.073)	0.201 (0.083)	0
Vasicek, Euler	0.158 (0.073)	0.199 (0.082)	0
CIR, exact	0.113 (0.060)	0.153 (0.070)	1/2
CIR, Euler	0.093 (0.060)	0.144 (0.069)	1/2
CKLS, Euler	0.080 (0.057)	0.144 (0.068)	0.77 (0.033) daily; 0.60 (0.149) monthly

- Data: daily DTB3 ($n = 13\,487$) and month-end values ($n = 647$)
- 2008–2026 is excluded: exact zeros and negative quotes lie outside the domain of these likelihoods (Appendix)
- Exact and Euler differ by less than one standard error: discretisation error is small next to the long-span uncertainty of $\hat{\kappa}$
- Volatility rises with the level: HAC Wald tests reject $\gamma = 0$ at both frequencies and $\gamma = 1/2$ with daily but not with monthly data (statistics: Appendix)

Nonparametric Drift and Diffusion (deeper)

- No functional form: kernel averages of the increments around each level x
  - $\Delta r_t = r_{t+\Delta} - r_t$ is the next increment and $w_t(x)$ the weight of day t, large when r_t is close to x
  - $\hat{a}(x) = \sum_t w_t(x)\,\Delta r_t/\Delta$, $\hat{b}^2(x) = \sum_t w_t(x)\,(\Delta r_t)^2/\Delta$, with Nadaraya–Watson weights $w_t(x) = K\left((r_t - x)/h\right) / \sum_s K\left((r_s - x)/h\right)$, Gaussian kernel K and bandwidth h Stanton (1997)
- $\mathbb{E}[\Delta r \mid r = x]/\Delta = a(x) + O(\Delta)$ and $\mathbb{E}[(\Delta r)^2 \mid r = x]/\Delta = b^2(x) + O(\Delta)$: exact only as $\Delta \to 0$
  - The diffusion converges at the infill rate, the drift only at the long-span rate
  - The results hold for non-stationary but recurrent processes Bandi & Phillips (2003)
- Caution: nonparametric specification tests reject far too often with persistent data Pritsker (1998)

Nonparametric Drift and Diffusion of the T-Bill Rate



- Data: daily DTB3, 1954–2007, bandwidth $h = 0.44$ percentage points (Silverman's rule, $h = 1.06\,\hat{s}\,n^{-1/5}$, with $\hat{s}$ the standard deviation of the rate)
- The 95% pointwise bands are heteroskedasticity-robust but ignore serial dependence
- Drift: zero lies inside the band at 97% of the levels
- Diffusion: the median relative SE is 8%, and $b(r)$ rises from 0.5% to 4.7% (log-slope 0.90)
- No parametric shape fits at every level
  - The CKLS curve lies inside the band at 35% of the levels, CIR at 7% and Vasicek at 5% (descriptive shares)

MFM_ch11_np_diffusion

Recap: What Frequency Buys and What It Cannot

- Formulas to keep
  - Infill: $\sum_k (\Delta X_k)^2 \to \int_0^T b_t^2 \, \mathrm{d}t$, so the diffusion coefficient is learned from frequent sampling
  - Long span: $I(m) = T/\sigma^2$ and $\mathrm{Var}(\hat{\kappa}) \approx 2\kappa/T$, so drift and speed are learned only from more years
  - Bias: $\hat{\kappa}$ exceeds κ by roughly $4/T$
- Check: the same 20 years are sampled daily instead of monthly (21 times more observations). Which standard error shrinks, that of $\hat{\sigma}$ or that of $\hat{\kappa}$?
  - Answer: that of $\hat{\sigma}$, by about $\sqrt{21} \approx 4.6$ times; that of $\hat{\kappa}$ barely moves, since $2\kappa/T$ does not depend on the step

Why Prices Use a Different Probability: a One-Period Example

- A stock costs 100 today and tomorrow is worth 120 (up) or 90 (down); the interest rate is 0; the real-world probability of up is $p = 0.6$
- A claim pays 1 if up and 0 if down; replicate it with Δ shares and B in cash
  - $120\Delta + B = 1$ and $90\Delta + B = 0$ give $\Delta = 1/30$, $B = -3$: the replica costs $1/3$, and any other price is an arbitrage
- The same price is the expectation under $q = (100 - 90)/(120 - 90) = 1/3$, the only weight with $120q + 90(1 - q) = 100$; p never enters
- In general, absence of arbitrage means that discounted traded prices are martingales under an equivalent measure Q (fundamental theorem of asset pricing, Shreve (2004), Ch. 5); Girsanov builds Q in continuous time
- Check: what is the price of the claim if $p = 0.9$? And if the down state is 80 instead of 90?
  - Answer: still $1/3$; with 80, $q = 20/40 = 1/2$, so the price is $1/2$

Girsanov's Theorem

- Real-world measure P and an equivalent measure Q (the same events of probability zero), linked by the **Radon–Nikodym density**
  - $\left.\frac{\mathrm{d}Q}{\mathrm{d}P}\right|_{\mathcal{F}_T} = Z_T = \exp\left(-\int_0^T \theta_s \,\mathrm{d}W_s - \frac{1}{2}\int_0^T \theta_s^2 \,\mathrm{d}s\right)$, with θ_s adapted (below: the market price of risk)
  - $Z_T > 0$ with $\mathbb{E}^P[Z_T] = 1$ reweights the probability of each path: $Q(A) = \mathbb{E}^P[Z_T 1_A]$
- Girsanov (1960): under Q, $\tilde{W}_t = W_t + \int_0^t \theta_s \,\mathrm{d}s$ is a Brownian motion if $\mathbb{E}\left[e^{\frac{1}{2}\int_0^T \theta_s^2 \mathrm{d}s}\right] < \infty$ (Novikov condition)
- For $\mathrm{d}X = a\,\mathrm{d}t + b\,\mathrm{d}W$: $\mathrm{d}X = (a - b\,\theta)\,\mathrm{d}t + b\,\mathrm{d}\tilde{W}$; for continuous Itô processes **only the drift changes**, while the diffusion and the quadratic variation are the same under P and Q
  - With jumps (Section 9), Q may also change the jump intensity and the jump-size distribution: $\hat{\lambda}$ from returns is not the intensity that prices options
  - With stochastic volatility, P and Q weigh future variance paths differently: historical variance is a benchmark, not a calibration

Market Price of Risk: P versus Q

- GBM: $\theta = (\mu - r)/\sigma$ is the **market price of risk**; under Q, $\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}\tilde{W}$ and $e^{-rt}S_t$ is a Q-martingale (Seminar 11, A4)
  - S&P 500, 1990–2026: $\hat{\theta} = (10.0\% - 2.7\%)/18.0\% = 0.40$, with r the mean 3-month T-bill rate; SE $1/\sqrt{T} = 0.17$, as imprecise as the drift
- Vasicek with risk premium $\lambda\sigma$: $\kappa^Q = \kappa$, $\kappa\theta^Q = \kappa\theta - \lambda\sigma$, with θ, θ^Q long-run levels
  - The long-run level that prices bonds differs from the one estimated on data (Section 6)
  - A positive term premium means $\lambda < 0$: then $\theta^Q = \theta - \lambda\sigma/\kappa > \theta$ and long yields are higher
- Heston with volatility risk premium λv_t (a separate λ): $\kappa^Q = \kappa + \lambda$, $\kappa^Q\theta^Q = \kappa\theta$
  - $\lambda < 0$ (investors pay for protection against variance) gives $\theta^Q > \theta$: the variance risk premium of Chapter 9

Feynman–Kac: From Expectations to the Pricing PDE

- If $\mathrm{d}X = a(X)\,\mathrm{d}t + b(X)\,\mathrm{d}\tilde{W}$ under Q Kac (1949)
  - $u(t,x) = \mathbb{E}^Q\left[e^{-\int_t^T r_s\,\mathrm{d}s}\, g(X_T) \mid X_t = x\right]$ solves the PDE (partial differential equation) $u_t + a\,u_x + \frac{1}{2}b^2 u_{xx} - r\,u = 0$, $u(T,x) = g(x)$
  - u is the price at t of the pay-off $g(X_T)$ at T, discounted at the short rate r_s
- Proof idea: $e^{-\int_0^t r_s\mathrm{d}s}\,u(t,X_t)$ is a Q-martingale, so its Itô drift is zero; by the same route, Chapter 12 derives Black–Scholes
- Vasicek bond ($g = 1$, $X = r$): $p(t,r;T) = \exp\left(A(\tau) - B(\tau)\,r\right)$, $\tau = T - t$ (Seminar 11, A6)
  - $B(\tau) = \frac{1-e^{-\kappa\tau}}{\kappa}$, $A(\tau) = \left(\theta^Q - \frac{\sigma^2}{2\kappa^2}\right)(B(\tau) - \tau) - \frac{\sigma^2 B(\tau)^2}{4\kappa}$
  - Yields $y(\tau) = (B(\tau)\,r - A(\tau))/\tau$ tend to $\theta^Q - \sigma^2/(2\kappa^2)$: rate uncertainty lowers long yields (convexity; comparison with Treasury yields: ▸ Appendix)

Recap: Pricing under Q

- Formulas to keep
  - Girsanov: under Q the drift a becomes $a - b\theta$, the diffusion b is unchanged
  - GBM: $\theta = (\mu - r)/\sigma$ turns the drift μ into r, so $e^{-rt}S_t$ is a Q-martingale
  - Feynman–Kac: price $= \mathbb{E}^Q$[discounted pay-off], the solution of the pricing PDE
- Check: $\mu = 9\%$, $\sigma = 20\%$, $r = 3\%$. What are the market price of risk and the drift of S under Q?
  - Answer: $\theta = 0.06/0.20 = 0.30$ and $\mu - \sigma\theta = 9\% - 6\% = 3\% = r$, with $\sigma = 20\%$ unchanged

Merton's Jump-Diffusion (1/2)

- **Question**: which model explains crash days that are practically impossible under GBM?
- Brownian paths are continuous, but prices gap on news: on 19 October 1987 the Dow Jones fell 22.6%, about 20 standard deviations under GBM with 18% volatility
  - A jump cannot be hedged by continuous trading Merton (1976)
- Log price = Brownian motion with drift + **compound Poisson** jumps: $\mathrm{d}\ln S_t = m\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + \mathrm{d}J_t$, $J_t = \sum_{i \le N_t} Y_i$
  - N_t is a Poisson process with intensity λ, and the jump sizes $Y_i \sim N(\mu_J, \sigma_J^2)$ are i.i.d.
  - W, N and Y are independent; the model has five parameters, $(m, \sigma, \lambda, \mu_J, \sigma_J)$

Merton's Jump-Diffusion (2/2)

- Given k jumps, the return over Δt is $N\left(m\Delta t + k\mu_J,\ \sigma^2\Delta t + k\sigma_J^2\right)$
- Unconditionally, it is a **Poisson mixture** of Normal densities, i.i.d. over time
  - $f(r) = \sum_{k\geq 0} e^{-\lambda\Delta t}\frac{(\lambda\Delta t)^k}{k!}\,\varphi\left(r;\ m\Delta t + k\mu_J,\ \sigma^2\Delta t + k\sigma_J^2\right)$
  - Weight: Poisson probability of k jumps; $\varphi(r;\text{mean},\text{variance})$: the Normal density
- Exact simulation: draw $N \sim \text{Poisson}(\lambda\Delta t)$, then draw from the Normal distribution with the conditional moments

Moments of the Merton Return

- Cumulants c_k of the return over Δt
  - Definition: $K(u) = \ln \mathbb{E}[e^{ur}]$ and $c_k = K^{(k)}(0)$: c_1 mean, c_2 variance, c_3 third central moment; cumulants of independent parts add, and a Normal part has $c_k = 0$ for $k \geq 3$
  - The k-th cumulant of a compound Poisson sum is $\lambda \Delta t \, \mathbb{E}[Y^k]$
  - $c_1 = m\Delta t + \lambda \Delta t \, \mu_J$, $c_2 = \sigma^2 \Delta t + \lambda \Delta t (\mu_J^2 + \sigma_J^2)$, $c_3 = \lambda \Delta t (\mu_J^3 + 3\mu_J \sigma_J^2)$, $c_4 = \lambda \Delta t (\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$
  - Skewness $c_3 / c_2^{3/2}$ is negative when $\mu_J < 0$; excess kurtosis c_4 / c_2^2 is positive with any jump
- Excess kurtosis falls like $1/\Delta t$ as the horizon grows: jumps matter most at short horizons
  - Example: with $\sigma = 15\%$, $\lambda = 5$, $\mu_J = -3\%$ and $\sigma_J = 4\%$, the daily excess kurtosis is 17.6 and the skewness -2.07 (Seminar 11, A3)

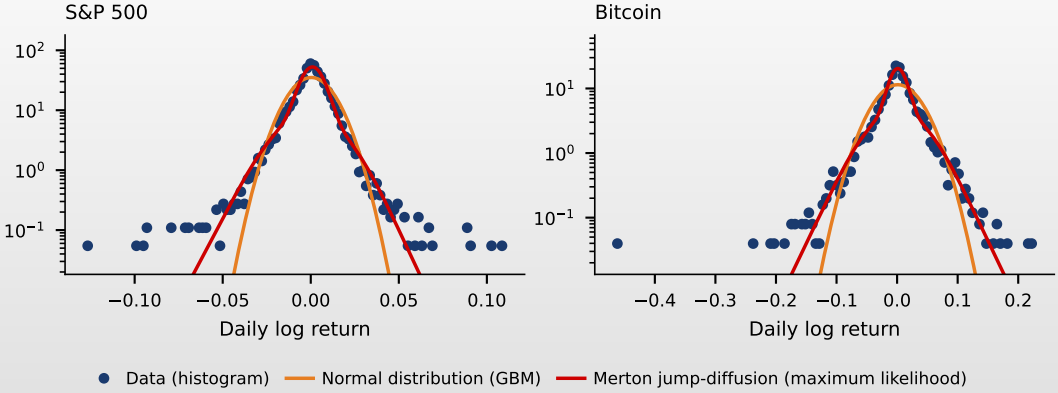
Merton Estimates for the S&P 500, BET and Bitcoin

LR (Likelihood Ratio): the likelihood-ratio statistic.

	$\hat{\sigma}$ (%)	$\hat{\lambda}$ (per year)	$\hat{\mu}_J$ (%)	$\hat{\sigma}_J$ (%)	Jump share of variance	LR
S&P 500	9.4	102 (11)	−0.16	1.44	71%	2 401
BET	11.3	79 (7)	−0.09	2.07	73%	2 326
Bitcoin	24.6	239 (21)	0.00	3.85	85%	1 357

- The Poisson-mixture log-likelihood (up to 12 jumps per day) is maximised numerically; in brackets, SE from the Hessian
- The likelihood is **unbounded**: with $m\Delta t$ equal to one observed return and $\sigma \to 0$ that density explodes, so the MLE is a local maximum Honoré (1998)
  - The LR test against GBM ($\lambda = 0$) is defined on $\sigma \geq 5\%$ and is not χ^2_3: $\lambda = 0$ lies on the boundary and μ_J, σ_J are unidentified (bootstrap critical values: Seminar 11, B5)
- The fit prefers **many small** jumps (about 80–240 per year), not a few crashes

Merton Fit to the Return Distribution (1/2)



MFM_ch11_merton_density

Merton Fit to the Return Distribution (2/2)

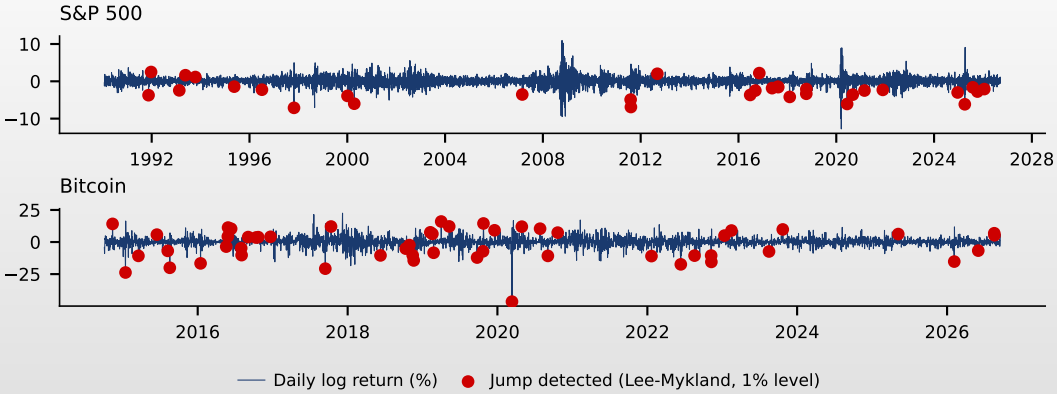
- Density of daily log returns on a log scale: histogram, the Normal distribution implied by GBM and the fitted Merton mixture
  - Merton follows the peak and the shoulders far better
- S&P 500: the Merton excess kurtosis is 3.7 against 10.9 in the data, and the skewness -0.31 against -0.36
- The most extreme days lie beyond both fitted curves
- About 102 "jumps" a year: the model uses jumps to mimic **changing volatility**
- An i.i.d. model cannot tell a jump from a volatile period, so a direct test is needed

MFM_ch11_merton_density

Detecting Jumps: the Lee–Mykland Test

- Idea: a jump is a return that is huge **relative to the local volatility** Lee & Mykland (2008)
  - $L_t = r_t/\hat{\sigma}_t$, $\hat{\sigma}_t^2 = \frac{1}{K-2}\sum_{j=t-K+2}^{t-1} |r_j||r_{j-1}|$ (bipower variation, robust to jumps)
  - With daily data the window is $K = 16$ days, as recommended by the authors
- Under no jumps, the maximum of $|L_t|$ over n days has a Gumbel limit
  - Reject if $(|L_t| - C_n)/S_n > -\ln(-\ln(1-\alpha))$, a threshold of 4.60 at $\alpha = 1\%$ for the **normalised** statistic, not for $|L_t|$
  - $C_n = \frac{\sqrt{2\ln n}}{c} - \frac{\ln\pi + \ln\ln n}{2c\sqrt{2\ln n}}$, $S_n = \frac{1}{c\sqrt{2\ln n}}$, $c = \sqrt{2/\pi}$ (the mean of $|Z|$, which rescales the bipower estimate), n = number of tested days
  - The test controls, asymptotically, the probability of a false jump over the **whole** sample, not day by day

Jumps Detected in the S&P 500 and Bitcoin (1/2)



MFM_ch11_jumps

Jumps Detected in the S&P 500 and Bitcoin (2/2)

	S&P 500	Bitcoin
Candidates	32	52
Per year	0.87	4.35
Negative (%)	84	52
In the top 10 moves	0	5

- Daily log returns; days flagged at the 1% level
- S&P 500: the largest moves (2008, 2020), which came in turbulent periods, are not flagged
  - Instead, a 1.2% move in a calm market is flagged (1993-10-14)
- Detections differ from the fitted Merton intensity. Possible explanations:
  - stochastic volatility
  - drift bursts (short explosive drift episodes)
  - low detection power in daily data

MFM_ch11_jumps

Recap: What Jumps Add

- Formulas to keep
  - $\mathrm{d}\ln S_t = m\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + \mathrm{d}J_t$: mean log return $m + \lambda\mu_J$ per year, so m alone is the drift of the diffusion part only
  - Higher cumulants come only from the jumps: $c_k = \lambda\Delta t\,\mathbb{E}[Y^k]$ for $k \geq 3$
  - Excess kurtosis c_4/c_2^2 falls like $1/\Delta t$ with the horizon
- From the data: S&P 500 $\hat{m} + \hat{\lambda}\hat{\mu}_J = 24.8\% + (-16.5\%) \approx 8.3\%$, the same as the GBM log drift
- Check: jumps give a daily excess kurtosis of 12. What is it for 10-day returns, if returns are i.i.d.?
  - Answer: $12/10 = 1.2$: over 10 days both c_4 and c_2 grow 10 times, so c_4/c_2^2 falls 10 times

The Heston Model

- **Question**: which model generates volatility clustering and negative skewness (the leverage effect), which GBM and Merton cannot produce?
- Two SDEs, for the log price and for the instantaneous variance v_t Heston (1993)
  - $\mathrm{d}\ln S_t = \left(\mu - \frac{1}{2}v_t\right)\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^{(1)}$
  - $\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^{(2)}$, $\mathrm{Corr}(\mathrm{d}W^{(1)}, \mathrm{d}W^{(2)}) = \rho$: the variance is a CIR process with level θ, speed κ and volatility of volatility ξ
  - Feller condition $2\kappa\theta \geq \xi^2$: the variance never touches zero Feller (1951)
- $\rho < 0$: the **leverage effect**; a price fall raises the variance that scales the next moves, so returns are negatively skewed (Section 11)
- Simulation: Euler scheme with full truncation $v^+ = \max(v, 0)$ Lord, Koekkoek & van Dijk (2010), or the quadratic-exponential scheme Andersen (2008)

Heston Parameters from the VIX

- Proxy $y_t = (\mathrm{VIX}_t/100)^2$ in place of v_t; discretised CIR regression without an extra intercept
  - $\frac{y_{t+\Delta t}-y_t}{\sqrt{y_t}} = \kappa\theta\frac{\Delta t}{\sqrt{y_t}} - \kappa\Delta t\sqrt{y_t} + \xi\sqrt{\Delta t}\,\varepsilon_t$
- S&P 500 and VIX, 1990–2026: $\hat{\kappa} = 4.13$ (half-life 42 days), $\hat{\xi} = 0.56$ and $\hat{\rho} = -0.69$
  - The long-run level from the VIX is $\sqrt{\hat{\theta}} = 20.9\%$, but the realised volatility is 18.0%
  - The Feller ratio of the proxy is $2\hat{\kappa}\hat{\theta}/\hat{\xi}^2 = 1.16$
- But y_t is **not** v_t: it is a risk-neutral expectation of future variance, affine in v_t, so $\hat{\xi}$ is attenuated (next slide)
  - For the simulated paths we set θ to the realised variance of the S&P 500

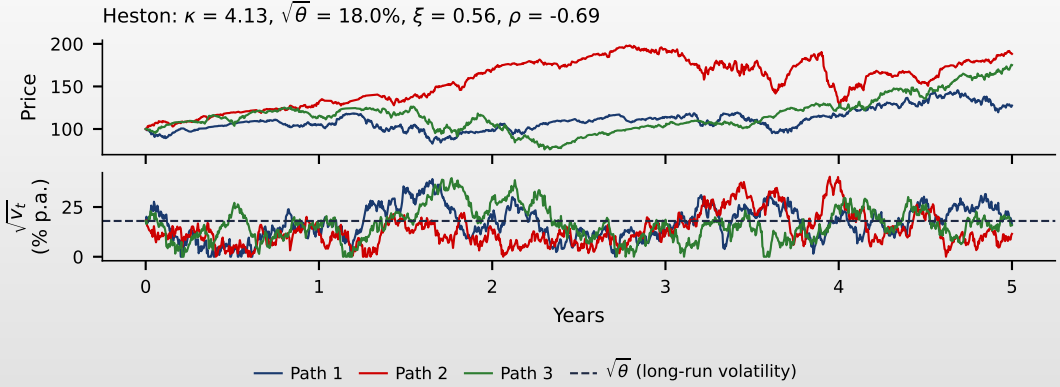
What the VIX Identifies in Heston (deeper)

- Under Q: $y_t = \frac{1}{\tau}\mathbb{E}^Q\left[\int_t^{t+\tau} v_s \, \mathrm{d}s \mid \mathcal{F}_t\right] = a + b\, v_t$, $\tau = 30/365$ (Seminar 11, A8)
  - $b = \frac{1-e^{-\kappa^Q \tau}}{\kappa^Q \tau} < 1$, $a = \theta^Q(1-b)$: y_t is **affine** in v_t, with Q-parameters
- What the regression on y_t recovers
  - κ: the P-speed, since an affine map keeps the dynamics of v_t
  - ρ: unchanged (correlation is scale-free), but only approximated by daily proxy changes
  - ξ is attenuated, and the long-run level mixes θ^Q and θ^P (derivation: next slide)
- Proper estimation fits the dynamics of y directly, with diffusion proportional to $\sqrt{y-a}$, or v_t and the observation equation jointly

Where the Attenuation of ξ Comes From (deeper)

- Step 1: $y_t = a + b\,v_t$ with constants a, b, so $\mathrm{d}y_t = b\,\mathrm{d}v_t$ under P
  - $\mathrm{d}y_t = b\kappa(\theta^P - v_t)\,\mathrm{d}t + b\,\xi\sqrt{v_t}\,\mathrm{d}W_t^{(2)}$
- Step 2, drift: $b\kappa(\theta^P - v_t) = \kappa(\theta_y - y_t)$ with $\theta_y = a + b\,\theta^P$: the speed κ is unchanged and $\theta^P = (\theta_y - a)/b$
- Step 3, diffusion: the CIR regression assumes $\xi_y\sqrt{y_t}$; matching $\xi_y\sqrt{y_t} = b\,\xi\sqrt{v_t}$ gives
  - $\xi_y = b\,\xi\sqrt{v_t/(a + b\,v_t)}$; at $v_t = \theta^Q$, $a + b\,\theta^Q = \theta^Q$, so $\xi_y = b\,\xi$
- Numbers: with $\kappa^Q = 5$, $b = 0.82$, so the fitted $\hat{\xi} = 0.56$ corresponds to $\hat{\xi}/b = 0.68$ near $v_t = \theta^Q$ (other κ^Q: Appendix; the correction $\hat{\xi}/b$: Appendix)
- κ^Q is not identified from the VIX alone: joint P/Q estimation with option prices Chernov & Ghysels (2000), or moments of realised variance Bollerslev & Zhou (2002)

Simulated Heston Paths (1/2)



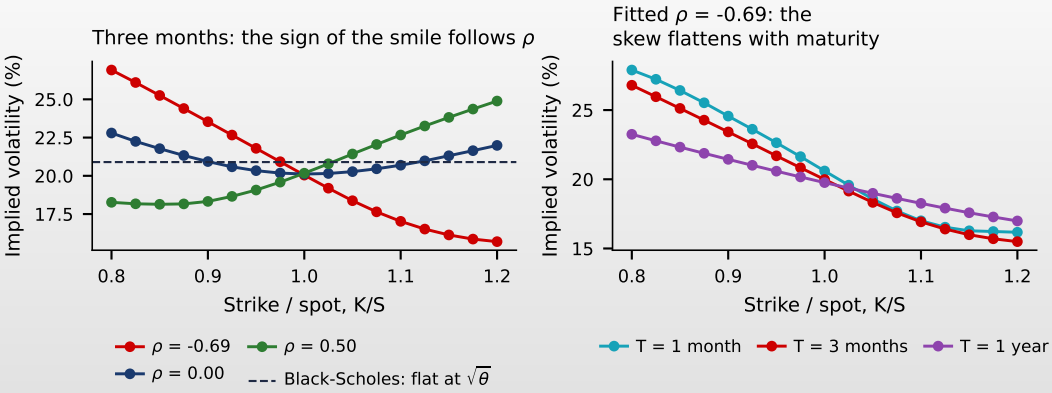
MFM_ch11_heston_paths

Simulated Heston Paths (2/2)

- Three five-year paths: top, the price; bottom, the volatility $\sqrt{v_t}$
  - $\hat{\kappa}$, $\hat{\xi}$ (uncorrected) and $\hat{\rho}$ come from the VIX, and θ is the realised S&P 500 variance
- Volatility clustering now comes from the model
  - Price falls coincide with volatility spikes
  - Simulated return–variance correlation: -0.69 (leverage through ρ)
- Feller ratio $2\hat{\kappa}\theta/\hat{\xi}^2 = 0.86 < 1$: the variance can reach zero
  - The negative values of the Euler scheme (1.4% of the steps) are set to zero (full truncation)

MFM_ch11_heston_paths

A Preview of Chapter 12: the Volatility Smile (1/2)



MFM_ch11_heston_smile

A Preview of Chapter 12: the Volatility Smile (2/2)

- **Implied volatility**: the σ that makes the Black–Scholes price equal to a given option price, flat across strikes under GBM; here from Heston call prices (Monte Carlo, 100,000 antithetic paths)
- Left, three months to maturity
  - with $\hat{\rho} = -0.69$, implied volatility is 26.9% at a strike of 80% of spot, 20.0% at the money and 15.7% at 120%: a downward **skew**
  - with $\rho = 0$, a symmetric **smile**: 22.8% at 80% of spot, 20.1% at the money and 22.0% at 120%
  - with $\rho = 0.50$, the skew reverses: implied volatility rises with the strike, from 18.3% to 24.9%
- Right: the skew flattens with maturity, from 27.9% – 16.2% at one month to 23.2% – 17.0% at one year
- Illustration only: pricing needs κ^Q, θ^Q calibrated to option prices (Chapter 12)

MFM_ch11_heston_smile

Affine Jump-Diffusions (deeper)

Affine jump-diffusions Duffie, Pan & Singleton (2000)

- Drift, variance, jump intensity affine in the state X_t
  - Vasicek, CIR, Merton, Heston and Bates
- $\mathbb{E}\left[e^{iu^\top X_T} \mid X_t\right] = \exp\left(\alpha(\tau) + \beta(\tau)^\top X_t\right)$
  - $i = \sqrt{-1}$, u: a real vector, $\tau = T - t$
  - α, β: solutions of Riccati ODEs
- Option prices by Fourier inversion
  - Simplest case: the Vasicek $A(\tau)$, $B(\tau)$ of Section 8

From GARCH to Diffusions (deeper)

GARCH (Generalised Autoregressive Conditional Heteroskedasticity) models conditional variance in discrete data.

From GARCH to continuous time Nelson (1990)

- As the step shrinks, GARCH(1,1) converges to
  - $\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi v_t\,\mathrm{d}W_t^{(2)}$
  - Like Heston, with v_t instead of $\sqrt{v_t}$
- $\kappa \approx (1-\alpha-\beta)/\Delta t$; α, β: GARCH(1,1) coefficients
  - Daily $\alpha+\beta = 0.99$: $\kappa \approx 2.5$ per year, half-life about 70 days (Chapter 5)
- GARCH and its diffusion limit are not asymptotically equivalent statistical experiments Wang (2002): inference can differ even as $\Delta t \to 0$

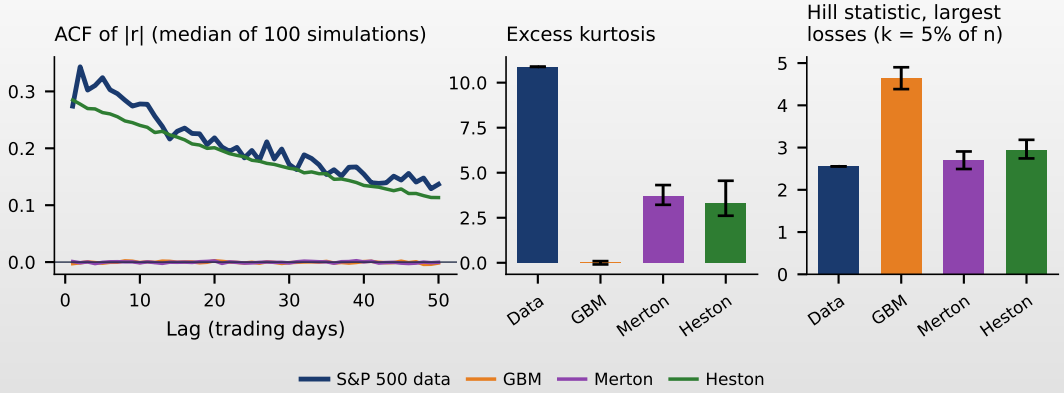
Recap: Stochastic Volatility

- Formulas to keep
  - Variance is a CIR process, $\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^{(2)}$, positive if $2\kappa\theta \geq \xi^2$
  - $\rho = \mathrm{Corr}(\mathrm{d}W^{(1)}, \mathrm{d}W^{(2)}) < 0$ gives leverage, negatively skewed returns and a downward implied-volatility skew
  - $(\mathrm{VIX}/100)^2 = a + b\,v_t$ with Q-parameters: κ and ρ carry over, ξ is attenuated by b
- Check: with $\kappa = 4$ and $\theta = 0.04$, what is the largest ξ for which the Feller condition holds?
  - Answer: $\xi \leq \sqrt{2 \cdot 4 \cdot 0.04} = \sqrt{0.32} \approx 0.57$; a larger vol-of-vol lets the variance touch zero

Three Yardsticks for Simulated Returns

- **Question**: which estimated model (GBM, Merton, Heston) reproduces the shape, the volatility clustering and the tail of S&P 500 returns?
- **Shape**: skewness $c_3/c_2^{3/2}$ and excess kurtosis c_4/c_2^2 of daily returns (cumulants: Section 9)
- **Clustering**: the ACF of $|r_t|$ at lags 1–20, zero for any i.i.d. model, Merton included
- **Tail**: Hill statistic of the k largest losses $L_{(1)} \geq \cdots \geq L_{(k+1)}$, $\hat{\alpha} = \left[\frac{1}{k}\sum_{i=1}^{k} \ln\left(L_{(i)}/L_{(k+1)}\right)\right]^{-1}$
  - It estimates α in $P(L > x) \approx Cx^{-\alpha}$ for large x: $P(L > 2x)/P(L > x) \approx 2^{-\alpha}$, so a lower α is a heavier tail
  - With $\alpha = 2.56$, doubling the loss leaves $2^{-2.56} \approx 0.17$ of the frequency; a Normal tail leaves $P(Z > 4)/P(Z > 2) \approx 0.0014$
  - The estimate depends on k: for the data, 2.97, 2.56 and 2.04 at $k = 2.5\%$, 5% and 10% of the returns

Which Model Reproduces the S&P 500? (1/2)



MFM_ch11_models_vs_data

Which Model Reproduces the S&P 500? (2/2)

- 100 simulated samples of 9 245 days from each fitted model (medians, 90% ranges)
  - Statistics compared: ACF of $|r_t|$, excess kurtosis (10.88 in the data) and the Hill statistic at $k = 5\%$ of the returns
- Merton gets the Hill statistic (2.70 against 2.56) and the skewness (-0.32 against -0.36), but no clustering
- Heston, with the VIX-based parameters, gets the clustering
- The simulated Heston skewness is 0.00 by construction: the simulation uses a one-day Euler step, while continuous-time Heston implies about -0.20 at these parameters (next slide)
- No fitted model reaches the kurtosis of the data (full table: Appendix)
  - Andersen, Benzoni & Lund (2002) and Eraker, Johannes & Polson (2003) favour jumps **and** stochastic volatility (SVJ, stochastic volatility with jumps, Bates (1996))

MFM_ch11_models_vs_data

Why a Negative ρ Skews Daily Returns (deeper)

- Over one day $h = 1/252$, ignoring drift terms of order h: $R = \int_0^h \sqrt{v_s}\,\mathrm{d}W_s^{(1)}$
  - Itô's lemma on $\sqrt{v}$: $\mathrm{d}\sqrt{v_s} = \frac{1}{2}\xi\,\mathrm{d}W_s^{(2)} + O(\mathrm{d}s)$, so $\sqrt{v_s} \approx \sqrt{v_0} + \frac{1}{2}\xi W_s^{(2)}$
  - $R \approx \sqrt{v_0}\,W_h^{(1)} + \frac{1}{2}\xi \int_0^h W_s^{(2)}\,\mathrm{d}W_s^{(1)}$
- Third moment: the cross term gives $\mathbb{E}[R^3] \approx \frac{3}{2}\,\xi\rho\,v_0 h^2$, while $\mathrm{Var}(R) \approx v_0 h$
  - Skewness $\approx \frac{3}{2}\,\rho\xi\sqrt{h}/\sqrt{v_0}$: with $\hat{\rho} = -0.69$, $\hat{\xi} = 0.56$ and $\sqrt{v_0} = 18.0\%$, about -0.20, a leading-order approximation

Which Model When? Features, Identification and Limits

Feature in the data	Model	What the data identify	Unresolved
Positive prices, $\sqrt{t}$ scaling	GBM	σ precisely; μ only over very long spans	fat tails, clustering
Mean reversion of rates and of the VIX	OU, Vasicek, CIR, CKLS	$b(r)$ precisely; κ biased upwards, needs a long span	one factor too simple for the VIX
Gaps, short-horizon fat tails	Merton	λ, σ_J only under i.i.d. returns; the jump share depends on the frequency	no clustering
Clustering, leverage, implied-volatility skew	Heston	κ, ρ from the VIX; ξ attenuated; $\theta^Q \neq \theta^P$	kurtosis of the data
All of the above	Bates (Heston + jumps)	returns and option surfaces jointly	latent states: filtering needed

- Roughness and long memory of volatility, next, are features none of these one-factor models produces

Background: Fractional Brownian Motion and the Hurst Exponent (deeper)

- **Question**: can one process describe volatility that is irregular at small scales and persistent at large scales (the VIX ACF, Section 6)?
- **Fractional Brownian motion** B^H, $0 < H < 1$: a Gaussian process with $B_0^H = 0$ and $\mathrm{Cov}(B_s^H, B_t^H) = \frac{1}{2}\left(s^{2H} + t^{2H} - |t-s|^{2H}\right)$
  - $H = 1/2$ gives $\min(s,t)$, i.e. Brownian motion; increments have variance $\mathbb{E}[(B_{t+h}^H - B_t^H)^2] = h^{2H}$
  - $H < 1/2$: successive increments are negatively correlated and paths are rougher than Brownian motion; $H > 1/2$: positively correlated, smoother
- **Fractional OU**: the OU equation driven by B^H instead of W; one H sets both ends of its ACF
  - Short lags: the variogram behaves like $h^{2H} = h^{2\alpha+1}$, so the roughness index is $\alpha = H - 1/2$ ($\alpha < 0$: rough)
  - Long lags: the ACF decays like $h^{2H-2} = h^{-\beta}$, so $\beta = 2 - 2H$ ($\beta < 1$: non-integrable ACF, long memory)

Research Question: Rough, Yet Persistent?

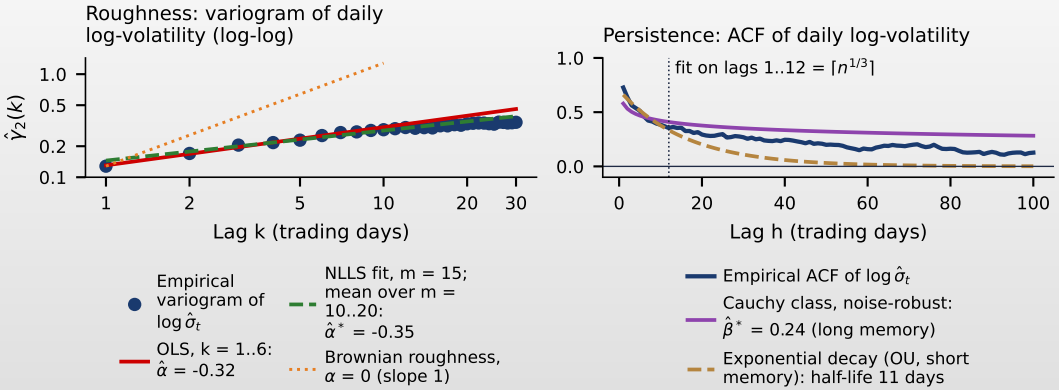
- Check: which H gives $\beta = 0.18$? Is that process rough?
  - Answer: $H = 1 - \beta/2 = 0.91$, so $\alpha = 0.41 > 0$: smooth, not rough; one H cannot match both features
- Paper: Bennedsen, Lunde & Pakkanen (2022), *Journal of Financial Econometrics*, which treats α and β as separate parameters
  - Builds on Gatheral, Jaisson & Rosenbaum (2018): log-volatility as a fractional OU process with $H < 1/2$ (RFSV, rough fractional stochastic volatility)
  - The long-memory model of Comte & Renault (1998) uses $H > 1/2$
- Question 1: are realised volatilities both rough and persistent?
- Question 2: do models that set α and β separately forecast better? (model, estimators and results: Appendix)

Case Study: Replication on SPY, $\Delta = 1$ Day (1/3)

- Data: SPY 5-minute bars, 1 468 full days (October 2020 – September 2026)
- $x_t = \log \hat{\sigma}_t = \frac{1}{2} \log \mathrm{BV}_t$, with BV_t the daily bipower variation of the 78 five-minute returns, robust to jumps Barndorff-Nielsen & Shephard (2004)
- Roughness: variogram $\hat{\gamma}_2(k) = \frac{1}{n-k} \sum_t (x_{t+k} - x_t)^2 \approx c\, k^{2\alpha+1}$ for small k
  - OLS: the slope $\hat{a}_1$ of $\log \hat{\gamma}_2(k)$ on $\log k$, $k = 1, \ldots, 6$, gives $\hat{\alpha} = (\hat{a}_1 - 1)/2$
  - Noise-robust NLLS: $\hat{\gamma}_2(k) = b_0 + b_1 k^{2\alpha+1}$, where b_0 absorbs the measurement noise in x_t
- Persistence: Cauchy-class ACF $\rho(h) = (1 + |h|^{2\alpha+1})^{-\beta/(2\alpha+1)}$, fitted on the first $\lceil n^{1/3} \rceil = 12$ lags
- Short-memory benchmark: an OU process, with ACF $e^{-\kappa h}$

Case Study: Replication on SPY, $\Delta = 1$ Day (2/3)

NLLS: Non-Linear Least Squares.



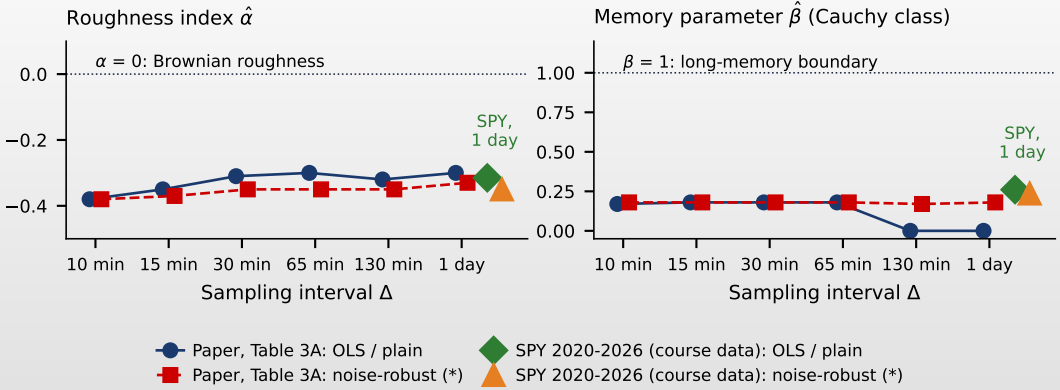
MFM_ch11_rough_vol

Case Study: Replication on SPY, $\Delta = 1$ Day (3/3)

- Left, roughness: empirical variogram $\hat{\gamma}_2(k)$ against k on log axes; the dotted line with slope 1 is Brownian roughness ($\alpha = 0$)
  - OLS gives $\hat{\alpha}_{\mathrm{OLS}} = -0.32$; the noise-robust NLLS gives $\hat{\alpha}^*_{\mathrm{NLLS}} = -0.35$
  - Bennedsen, Lunde & Pakkanen (2022) at 1 day: -0.30 (OLS) and -0.33 (NLLS)
- Right, persistence: the autocorrelation is 0.13 at lag 100
  - The Cauchy fit on 12 lags gives $\hat{\beta} = 0.26$ and $\hat{\beta}^* = 0.24$ (noise-robust), both below 1: consistent with long memory, not a proof
  - An OU fit (half-life 11 days) implies an autocorrelation of 0.002 at lag 100
- In the plot, m is the number of lags used in NLLS: 15 for the illustrated fit; the reported estimate averages fits over 10–20 lags.

MFM_ch11_rough_vol

Case Study: SPY Against Table 3 of the Paper (1/2)



MFM_ch11_rough_vol

Case Study: SPY Against Table 3 of the Paper (2/2)

- Lines: E-mini S&P 500 futures, 2011–2014, Δ from 10 minutes to 1 day (Bennedsen, Lunde & Pakkanen (2022), Table 3, Panel A)
- Markers: SPY at 1 day, 2020–2026, course data; * marks the noise-robust estimators
- $\hat{\alpha}$ barely moves with Δ (-0.38 to -0.30): roughness is a property of the process, and SPY falls in the same band
- $\hat{\beta}^* = 0.17$–0.18 at every Δ, far below 1, while the plain $\hat{\beta}$ is unstable
- Limits: another contract, period and estimator (bars, not every trade; no pre-averaging)
- Realised measures estimate integrated, not spot, variance

MFM_ch11_rough_vol

An Open Question: How Much of the Price Path Is Discontinuous?

- Model: $\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t + J_t\,\mathrm{d}N_t$, with X_t the log price, N_t the number of jumps (intensity λ) and J_t the jump size
- The answer depends on the sampling frequency and on the model
  - Daily Merton MLE (this chapter): jumps carry 71% of S&P 500 variance
  - The 5-minute bipower test (Chapter 9): jumps carry 2.8% of SPY variance
  - Christensen, Oomen & Podolskij (2014): at tick frequency most jumps detected at coarser sampling disappear; they are rapid but continuous moves
  - Christensen, Oomen & Renò (2022): **drift bursts**, short explosive trends in μ_t, look like jumps in coarse data
- Formal question: H_0: $\lambda = 0$ (continuous path with stochastic volatility) against H_1: $\lambda > 0$, a non-standard test with the boundary problem of the Merton LR test (Section 9)

The Discovery Loop with an AI Assistant

The loop with an AI (Artificial Intelligence) assistant

Assistant: an LLM (Large Language Model) such as Claude, ChatGPT or Gemini.

- **1. Literature**: published jump shares by frequency; check every Digital Object Identifier (DOI) on Crossref
- **2. Hypotheses**: Merton jumps, Heston volatility, drift bursts, each with a distinct testable implication
- **3. Replicate first** the chapter's estimates
- **4. Robustness**: filters and tests fixed before the first run
- **5. Critique**: the assistant challenges the inference, as a referee would
- **6. Report**: every specification, the code, the AI log
- Reusable prompts: ▸ Appendix

What the Researcher Checks: Failure and Test

Checks: failure and test

- Boundary, unidentified parameters: parametric bootstrap under H_0
- Look-ahead in the volatility filter: standardise only with $\mathcal{F}_{t-1}$
- Frequency mismatch: jump shares at every frequency on a common sample
- Multimodal likelihood: many starting points and a profile in λ
- Invented numbers: every claim needs a table and a DOI

Mini-Case: Jumps or Stochastic Volatility in Disguise?

- A typical assistant proposal: "71% of variance comes from jumps"
  - Rival hypothesis: the Poisson mixture absorbs time-varying volatility; then standardising by EWMA volatility (Exponentially Weighted Moving Average; $\lambda_{\mathrm{EWMA}} = 0.94$, the RiskMetrics value for daily data, known at $t-1$) removes the jumps

	Raw returns	EWMA-standardised
Excess kurtosis	10.9	2.9
LR, Merton vs the Normal distribution	2 401	619
$\hat{\lambda}$ per year (SE)	102 (11)	124 (53)
Jump share of variance	71%	51%

- Standardising removes most kurtosis, yet the model still reports a "jump" about every other day: $\hat{\lambda}$ measures fat tails, not discontinuities
- Interpretation: daily data do not identify the jump share

Your Turn: A Project Seed

- Questions
  - how much of S&P 500 and Bitcoin variance is discontinuous?
  - how does this share depend on the sampling frequency?
  - are the detected "jumps" drift bursts?
- Data: SPY 5-minute bars (October 2020 – September 2026) for every intraday frequency; S&P 500 and Bitcoin daily closes
- The course data contain no intraday Bitcoin series: an intraday comparison needs an external source documented in the pre-registration
- A pre-registration (hypotheses, frequencies, filters, tests, bootstrap design) dated before the first run

Your Turn: Replication and Core Deliverables

- Replicate first
  - the Merton jump share of variance: 71% (daily S&P 500)
  - the number of Lee–Mykland jumps: 0.87 per year
  - the bipower jump share: 2.8% (SPY, 5 minutes, Chapter 9)
- Core deliverables, feasible with the course data
  - SPY: bipower jump share and Lee–Mykland counts at 5, 15 and 30 minutes
  - The EWMA standardisation of the mini-case, repeated on SPY at each frequency

Your Turn: Event Window and Report

- The drift-burst statistic of Christensen, Oomen & Renò (2022) on SPY over 2–11 April 2025 (US tariff announcement on 2 April, 90-day pause on 9 April), against a calm window of equal length fixed in advance
- Optional extension: a bootstrap LR test of Heston (null, no jumps) against Bates
  - Both likelihoods need filtering of the latent variance, e.g. the Markov Chain Monte Carlo (MCMC) approach of Eraker, Johannes & Polson (2003); plan the computing time before committing to it
- Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Declare your AI use in `AI_USE.md` and the errors you caught in `AI_ERRORS.md`: github.com/danpele/MFM/tree/main/project

Key Takeaways and Next: Chapter 12

- Rescaled random walks converge to Brownian motion (Donsker), whose quadratic variation t drives Itô calculus
- Itô's lemma gives exact solutions (GBM, Ornstein–Uhlenbeck, Merton); other models are simulated with Euler–Maruyama or Milstein at a known error order
- Frequent sampling pins down volatility; drift and mean-reversion speed need long spans, and $\hat{\kappa}$ is biased upwards
- No arbitrage makes prices discounted Q-expectations; Girsanov changes only the drift, and Feynman–Kac turns the expectation into a PDE
- $(\mathrm{VIX}/100)^2$ is a Q-expectation, affine in v_t
- In the fitted models jumps help with the tails and stochastic volatility with the clustering; volatility looks rough at short lags and persistent at long lags, which a one-factor OU model does not reproduce
- Next, Chapter 12: pricing options by replication and reading the volatility surface

Check Your Understanding

- Why does a change of variable need an extra term in Itô calculus? Identify the role of quadratic variation.
- Does observing prices more frequently make the drift estimate precise? Separate frequency from the length of the observation period.
- Which feature calls for jumps, and which calls for stochastic volatility? Use the comparison with observed returns.
- Why can a pricing probability differ from a real-world probability? State what changes under the change of measure.

From Brown to Bachelier, Einstein and Wiener



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Albert Einstein in 1905

Photo: Lucien Chavan / ETH Zürich (1905); public domain; Wikimedia Commons

- 1827: the botanist Robert Brown sees pollen particles move in an irregular zig-zag under the microscope
- Bachelier (1900): French government bond (rente) prices and their options follow a random walk in continuous time
  - The spread of the price grows with $\sqrt{t}$, five years before physics
- Einstein (1905): molecular collisions explain Brown's motion; the mean squared displacement is linear in t
- Wiener (1923): Brownian motion as a probability measure on continuous paths, the **Wiener process**

Back

Itô, Samuelson, Black, Scholes and Merton



Kiyosi Itô (1915–2008)

Photo: Konrad Jacobs (1970); CC BY-SA 2.0 de; Wikimedia Commons



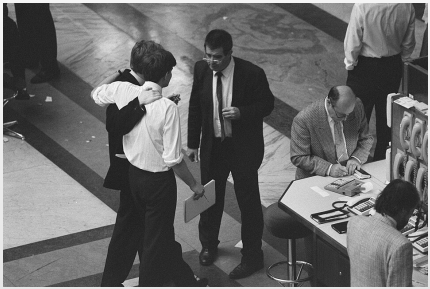
Paul A. Samuelson (1915–2009)

Photo: Bernard Gotfryd (1970–1975); public domain; Wikimedia Commons

- Itô (1944): the stochastic integral and its chain rule, the main working tool of this chapter
- Osborne (1959): log prices, not prices, behave like Brownian motion; Samuelson (1965): geometric Brownian motion
- Black & Scholes (1973) and Merton (1973): an option can be replicated in continuous time
- Merton (1976) adds jumps, Heston (1993) random volatility; textbook treatment: Franke, Härdle & Hafner (2019), Part I

Back

A Crash in Pictures, Amsterdam 1987



Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- The fall of 19 October 1987 (Section 9): 22.6% in one day
- Under a Normal distribution such a day would not be expected once in the age of the universe: prices gap, and continuous models need jumps or fat-tailed innovations

Back

Data Cleaning and Technical Conditions

- Data: equity indices on weekdays, stale quotes removed (unchanged close and no intraday range), genuine zero returns kept; Bitcoin 7 days a week
- Itô integral: left-point sums converge for regular integrands (e.g. continuous and adapted); a general H is reached by approximation with simple processes
  - If only $\int_0^T H_t^2 \, \mathrm{d}t < \infty$ almost surely, $\int H \, \mathrm{d}W$ is merely a local martingale and its mean need not be zero
- SDE: existence and uniqueness of a strong solution if a and b are Lipschitz in x with at most linear growth Kloeden & Platen (1992)
  - Square-root coefficients (CIR, Heston) are not Lipschitz at zero, yet have unique strong solutions (Yamada–Watanabe conditions)
- Donsker: convergence in distribution in the space of càdlàg paths with the Skorokhod topology Billingsley (1999)

Back

Short Rates, More Numbers

- DTB3, 1954–2026: the rate ranged from -0.05% (2020-03-26) to 17.1% (1980-12-11); 4.3% of the days were below 0.05%
  - The fitted Vasicek model gives a 6.1% long-run probability of a negative rate
  - 2008–2026: 13 exact zeros and 8 negative quotes, outside the domain of the CIR and CKLS likelihoods; 17% of the days form a near-zero regime below 0.05%
- CKLS tests on γ, 1954–2007, HAC Wald with 11 (daily) and 6 (monthly) lags
  - $\gamma = 0$ rejected: daily 545, monthly 16; $\gamma = 1/2$ rejected daily (67.0; 22.1 with 55 lags), not monthly (0.5 against 3.84)
  - The Gaussian LR for $\gamma = 1/2$ rejects monthly (10.1): it ignores volatility clustering; Chan, Karolyi, Longstaff & Sanders (1992) found $\hat{\gamma} \approx 1.5$ for 1964–1989

Back

Vasicek Yield and the Observed Rate

- Vasicek bond with the Section 6 estimates, $\lambda = 0$ and $r = 3.99\%$: 10-year yield 4.05%; the 10-year Treasury yield (FRED series DGS10) on 2026-09-18: 5.01%
  - Conventions first: DGS10 is a semiannual par yield, $2\ln(1 + 5.01\%/2) = 4.95\%$ continuously compounded; DTB3 is a 91-day discount rate, 3.99% is 4.07% continuously compounded and lifts the model yield to 4.10%
  - The gap shrinks only from 0.96 to 0.85 percentage points (par versus zero-coupon yield is not adjusted): it is not a quoting artefact, but a term premium ($\lambda \neq 0$; matching it needs $\theta^Q > \hat{\theta}$, i.e. $\lambda < 0$) plus misspecification
- Nonparametric estimation: Stanton (1997) also estimates the market price of interest-rate risk; specification test: Aït-Sahalia (1996)

Back

The VIX Correction by κ^Q and the Full Model Comparison

κ^Q	1	2.5	5	10
b	0.96	0.90	0.82	0.68
Local correction $\hat{\xi}/b$	0.58	0.62	0.68	0.82

	Data	GBM	Merton	Heston
Excess kurtosis	10.88	0.00	3.66	3.30
Skewness	−0.36	0.00	−0.32	0.00†
Hill statistic	2.56	4.64	2.70	2.95
ACF of $\lvert r\rvert$, lag 1	0.27	0.00	0.00	0.28
ACF of $\lvert r\rvert$, sum of lags 1–20	5.32	0.00	0.00	4.78

- $\hat{\xi}/b$ is exact only at $v_t = \theta^Q$. Model columns: medians of 100 simulations
- A Heston model with a lower Feller ratio could reach the kurtosis: short-horizon excess kurtosis $\approx 3\xi^2/(2\kappa\theta)$
- † Zero by construction with the one-day Euler step (Section 11)

Back

Further Study: Reading

- Heavier tails: Lévy processes and α-stable distributions Cont & Tankov (2004)
- Stochastic calculus: Shreve (2004), Ch. 5–6; Franke, Härdle & Hafner (2019), chapters on the Wiener process and stochastic integration
- Exercises with solutions on all models of this chapter: Borak, Härdle & López-Cabrera (2013); the Heston model: Čížek, Härdle & Weron (2011); applications: Härdle, Chen & Overbeck (2017)
- Monte Carlo methods: Glasserman (2003); numerical SDEs: Kloeden & Platen (1992)

Back

Case Study, Model, Data and Method (1/3)

- **Model**: $\sigma_t = \xi e^{X_t}$ (eq. (1.3)), $X_t = \int_{-\infty}^{t} g(t-s)\, v_s \, \mathrm{d}W_s$, a Brownian semistationary (BSS) process with kernel g and volatility-of-volatility process v_s (eq. (1.5)); the forecasts take $v_s \equiv 1$, a Gaussian X (Section 5)
  - Gamma kernel $g(x) = x^{\alpha} e^{-\lambda x}$ (Example 1.2): roughness α, short memory with exponential decay at rate λ
  - Power kernel $g(x) = x^{\alpha}(1+x)^{-\gamma-\alpha}$ (eq. (1.8)): $\beta = \gamma$ if $\gamma > 1$, $\beta = 2\gamma - 1$ if $\gamma \in (1/2, 1)$ (Remark 1.2); a BSS process with this kernel is the Power-BSS model
- Gaussian alternative: the Cauchy class (Section 1.1.1)
  - $\rho(h) = (1 + |h|^{2\alpha+1})^{-\beta/(2\alpha+1)}$: roughness α and memory β set separately

Back

Case Study, Model, Data and Method (2/3)

- **Step 1**: α from the variogram of $\log \hat{\sigma}$, by OLS (eq. (2.1)) and NLLS as in the SPY replication; the NLLS estimate averages $m = 10, \ldots, 20$ lags (Sections 2.1 and 3.1)
- **Step 2**: β, γ or λ by matching the model ACF to the empirical one on the first $\lceil n^{1/3} \rceil$ lags (Sections 2.2 and 4.1)
- **Data**: E-mini S&P 500 futures, every trade, 2011–2014 (996 days), 09:30–16:00 New York time
- $\hat{\sigma}_t^2$: pre-averaged bipower variation over Δ, divided by Δ, for Δ from 10 minutes to 1 day (Sections 4 and 4.1)
  - Bipower variation of the M returns r_j in a bin: $\mathrm{BV} = \frac{\pi}{2} \sum_{j=2}^{M} |r_j| \, |r_{j-1}|$, robust to jumps Barndorff-Nielsen & Shephard (2004); pre-averaging first averages returns over short windows to damp microstructure noise (Appendix A of the paper)
  - Intraday seasonality removed with the flexible Fourier form of Andersen & Bollerslev (1997)
- Daily data on 1 944 liquid US stocks, 2003–2013 (Sections 4.1 and 4.2)

Back

Case Study, Model, Data and Method (3/3)

- **Forecasts**: variance summed over the next $h = 1, 2, 5, 10, 20$ steps, rolling window of 200 observations, out of sample 2 January 2013 – 31 December 2014 (Section 5.1.1)
  - Rivals: random walk, AR(5), AR(10), ARMA(1,1), log-HAR (heterogeneous autoregressive model for log volatility) Corsi (2009), ARFIMA(0,d,0), RFSV
- **Forecast object**: $\mathrm{FO}_t(\Delta, h) = \sum_{k=1}^{h} \mathrm{BV}_{t+k\Delta}$ (eq. (5.3)); for a Gaussian log model, $\mathbb{E}[e^{X_{t+\Delta}} \mid \mathcal{F}_t] = \exp(m + s^2/2)$, with m, s^2 the conditional mean and variance (eq. (5.2))
- Losses over the out-of-sample origins: $\mathrm{MSE} = \mathrm{mean}\,(\widehat{\mathrm{FO}}_t - \mathrm{FO}_t)^2$, $\mathrm{QLIKE} = \mathrm{mean}\,(\log \widehat{\mathrm{FO}}_t + \mathrm{FO}_t / \widehat{\mathrm{FO}}_t)$
- **Model confidence set** (MCS) Hansen, Lunde & Nason (2011): a set of models that contains the best model with probability 90% (or 75%)
  - Membership is not a ranking: the models inside the MCS cannot be told apart at that level
- Scaling BV by $M/(M-1)$ (Seminar 11, C3) shifts x_t by $\frac{1}{2}\ln(78/77) \approx 0.006$ for $M = 78$, a constant that cancels in $\hat{\gamma}_2(k)$ and in the ACF

Back

Case Study, Results in the Paper

$\log\sigma$, E-mini S&P 500 (Table 3, Panel A)	$\hat{\alpha}_{\mathrm{OLS}}$	$\hat{\alpha}^*_{\mathrm{NLLS}}$	$\hat{\beta}^*_{\mathrm{Cauchy}}$	$\hat{\lambda}^*_{\mathrm{Gamma}}$
$\Delta = 10$ minutes	-0.38	-0.38	0.18	0.02
$\Delta = 65$ minutes	-0.30	-0.35	0.18	0.02
$\Delta = 1$ day	-0.30	-0.33	0.18	0.02

- Rough at every scale ($\hat{\alpha} < 0$, far rougher than Brownian motion, $\alpha = 0$) and persistent ($\hat{\beta}^* < 1$): long memory together with roughness; * = noise-robust; SPY comparison in the main text
  - 1 944 US stocks: $\hat{\alpha}^*$ mostly between -0.45 and -0.30, $\hat{\beta}^*$ mostly between 0 and 0.4, no systematic difference across sectors (Figures 4 and 5): both features look universal
- Noise can fake roughness: a simulated $\alpha = 0$ observed with noise-to-signal ratio 0.3 at $\Delta = 0.1$ gives a mean OLS estimate of -0.47 but a mean NLLS estimate of 0.01 (Table 1, Panel C)
- Forecasts: modelling persistence pays intraday and at long horizons
  - 15-minute steps, $h = 20$: MSE ($\times 10^{\mathbf{8}}$) 0.038 for log-HAR, 0.020 for RFSV, 0.019 for Power-BSS (Table 4, Panel A)
  - Daily steps, $h = 1$: RFSV is best, 0.493 ($\times 10^{\mathbf{9}}$) (Table 5, Panel B)

Back

Case Study, Intraday Application, Limits and Open Questions (1/3)

- **Intraday application** (illustrative, not an exact replication) on the SPY 5-minute bars of the course data
  - $\hat{\sigma}_t^2$ = bipower variation of the 5-minute returns in each bin of $\Delta = 30$ or 65 minutes
  - Divided by Δ and by the flexible Fourier seasonal factor Andersen & Bollerslev (1997)
  - OLS and NLLS $\hat{\alpha}$, Cauchy $\hat{\beta}^*$ and Gamma-BSS $\hat{\lambda}^*$ against Table 3, Panel A; log-HAR, RFSV and BSS forecasts, MSE, QLIKE and MCS against Table 5
  - Data limit: bars instead of every trade, so the pre-averaged estimator (Appendix A of the paper) is not feasible and the noise is larger; only 13 returns per 65-minute bin

Back

Case Study, Intraday Application, Limits and Open Questions (2/3)

- **Limits stated by the authors**
  - Realised measures are noisy estimates of integrated variance: the conclusions concern them, not the latent spot volatility (Sections 4.1, 4.2 and 6)
  - No rule for the NLLS bandwidth m (Sections 2.1 and 3.1); NLLS is too costly for rolling forecasts, which use the OLS $\hat{\alpha}$ (Section 5.1.1)
  - Forecasts assume a Gaussian X; a non-Gaussian volatility of volatility is left out of scope (Section 5; footnote 3)

Back

Case Study, Intraday Application, Limits and Open Questions (3/3)

- **Open question**: why does long memory improve even short-horizon forecasts? The authors leave it to further research (Section 5.1.2)
  - Rolling $\hat{\alpha}$ (65 minutes, 2005–2014, median -0.36) is less rough in 2007–2008 and 2011 (Figure 3)
  - $\hat{\alpha}$ rises with log volume, slope 0.0091 (Figure 6)
  - Does Bitcoin (5-minute bars), a market that never closes, fit this pattern?

Back

The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: Semantic Scholar, Elicit, Google Scholar
- **1. Literature**: tabulate estimates by frequency, then check every DOI on Crossref (the title must match)
  - `Tabulate published estimates of the jump share of equity-index variance by sampling frequency (daily, 5-minute, tick) and method; give the exact number, the table it comes from, and the DOI.`
- **2. Competing hypotheses**, with distinct testable implications
  - `For three processes that give a daily excess kurtosis near 11 (Merton jumps, Heston volatility, drift bursts), derive the implications for returns standardised by ex-ante volatility, for bipower versus realised variance, and for the drift just before extreme moves.`
- **3. Data and code**: replicate the chapter's estimate first
  - `Write Python that reads the S&P 500 daily closes of the course and fits the Merton model by maximum likelihood (Poisson mixture truncated at 12 jumps, several starting points).`

Back

The Discovery Loop, Reusable Prompts (2/2)

- **4. Robustness**, pre-registered
  - `Write a pre-registered plan: standardise returns by EWMA and by GARCH volatility known at t-1; split 1990-2007 and 2008-2026; test Heston (no jumps, the null) against Bates (SVJ); obtain LR critical values by parametric bootstrap under the Heston null.`
- **5. Critique**: ask the assistant to attack the inference
  - `Act as a referee: why is the LR test of lambda = 0 non-standard, what is the correct reference distribution, and could a Poisson mixture with frequent small jumps simply be a fat-tailed innovation?`
- **6. Report**: all frequencies and filters, bootstrap critical values, code and the AI log

Back

Further Study on Quantinar

Advanced courses on quantinar.com

- Applied Computational Finance
- XFG Advanced Methods in Quantitative Finance

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

Back

Acronyms Used in This Material (1/4)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BSS**: Brownian SemiStationary (process)
- **BV**: Bipower Variation
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CBOE**: Chicago Board Options Exchange
- **CI**: Confidence Interval
- **CIR**: Cox–Ingersoll–Ross (model)

Acronyms Used in This Material (2/4)

- **CKLS**: Chan–Karolyi–Longstaff–Sanders (model)
- **CLT**: Central Limit Theorem
- **DGS10**: 10-Year Treasury Constant Maturity Rate (FRED series code)
- **DOI**: Digital Object Identifier
- **DTB3**: 3-Month Treasury Bill: secondary market rate (FRED series code)
- **ETF**: Exchange-Traded Fund
- **EWMA**: Exponentially Weighted Moving Average
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GBM**: Geometric Brownian Motion
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)

Acronyms Used in This Material (3/4)

- **JB**: Jarque–Bera (test)
- **LLM**: Large Language Model
- **LR**: Likelihood Ratio (test)
- **MCMC**: Markov Chain Monte Carlo
- **MCS**: Model Confidence Set
- **MLE**: Maximum Likelihood Estimation
- **MSE**: Mean Squared Error
- **NLLS**: Non-Linear Least Squares
- **ODE**: Ordinary Differential Equation
- **OLS**: Ordinary Least Squares
- **OU**: Ornstein–Uhlenbeck (process)
- **PDE**: Partial Differential Equation

Acronyms Used in This Material (4/4)

- **QLIKE**: Quasi-LIKElihood (loss function)
- **QQ**: Quantile–Quantile (plot)
- **RFSV**: Rough Fractional Stochastic Volatility (model)
- **SDE**: Stochastic Differential Equation
- **SE**: Standard Error
- **SPY**: ticker of the SPDR S&P 500 ETF Trust
- **SVJ**: Stochastic Volatility with Jumps (model)
- **US**: United States
- **VaR**: Value-at-Risk
- **VIX**: Cboe Volatility Index
- **XFG**: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

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